

Representation Theory Final Exam Study Guide

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Contents

1	Things to Study	1
2	Final Exam Review	2
2.1	Theorems and Proofs of Theorems	2
2.2	Important Definitions for The Final Exam	8
2.2.1	Lecture 1: Foundations	10
2.2.2	Lecture 2: Modules and Group Algebra	11
2.2.3	Lecture 3: Reducibility and Schur	13
2.2.4	Lecture 4: Maschke and Structure	14
2.2.5	Lecture 5: Characters	15
2.2.6	Lecture 6: Orthogonality (Conceptual Only)	16
2.3	Will's Answers for Homework 3	16
2.4	Gonzalez's Answers for Homework 3	25
3	Test 2 Review	39
3.1	Theorems and Proofs of Theorems	39
3.2	Important Definitions for Test 2	42
3.3	Alternative Answers For Test 2	43
3.4	Test 2 Solutions	44
4	Test 1 Review	54
4.1	Alternative Answers for Test 1	54
4.2	Test 1 Solutions	55

Section 1: Things to Study

Note 1 (Final Exam Contents According to Sakai). The contents of the final exam will follow these topics:

1. Group Actions
2. Classification of Permutation Representations
3. Burnside's Lemma
4. Lectures 1-6 of Representation Theory (Definitions, Examples, Propositions, Lemmas, Corollaries)
5. Lecture 3: Thm 4.1
6. Lecture 4: Prop 2.6, Thm 3.2 (please read this proof, but it is not included in the final).
7. Lecture 5: Prop 1.1, Prop 3.2, Prop 4.1.(i)
8. Lecture 6: Prop 2.3, Prop 2.6, Cor 2.7, Thm 5.1
9. HWK 3

10. Previous two exams (I will choose at least one problem from one of the previous midterms)
(TEST 1 AND TEST 2)

Section 2: Final Exam Review

2.1 Theorems and Proofs of Theorems

Theorem 2.1 (Theorem L3-4.1: Irreducible representations of abelian groups). *Let G be a finite abelian group and V an irreducible representation of G over $\mathbb{C}$. Then*

$$\dim V = 1. \quad (2.1)$$

Proof. Let $\rho : G \rightarrow \text{GL}(V)$ be an irreducible representation and let $h \in G$. Then $\rho(h) : V \rightarrow V$ is a linear isomorphism. We claim $\rho(h)$ is a G -module homomorphism.

Indeed, for all $g \in G$ and $v \in V$:

$$\rho(h)(\rho(g)(v)) = \rho(hg)(v) \quad (2.2)$$

$$= \rho(gh)(v) \quad (2.3)$$

$$= \rho(g)(\rho(h)(v)), \quad (2.4)$$

where the middle equality uses the fact that G is abelian.

So $\rho(h)$ is a G -endomorphism of the irreducible module V .

By Schur's Lemma (ii),

$$\rho(h) = \lambda_h \cdot \text{id}_V \quad (2.5)$$

for some $\lambda_h \in \mathbb{C}$ (with $\lambda_h \neq 0$ since $\rho(h)$ is invertible).

Now fix any nonzero $w \in V$. For every $g \in G$,

$$g \cdot w = \lambda_g w \in \mathbb{C}\{w\}. \quad (2.6)$$

So $W = \mathbb{C}\{w\}$ is a one-dimensional G -submodule of V . Since V is irreducible and $W \neq \{0\}$, we get $V = W$, i.e.

$$\dim V = 1. \quad (2.7)$$

□

Proposition 2.2 (Proposition L4-2.6: Orthogonal complements of G -submodules). *Let V be a G -module with a G -invariant inner product $\langle \cdot, \cdot \rangle$, and let $W \subseteq V$ be a G -submodule. Then $W^\perp$ is also a G -submodule, and consequently*

$$V = W \oplus_G W^\perp. \quad (2.8)$$

Proof. Let $u \in W^\perp$ and $g \in G$. We must show $g \cdot u \in W^\perp$, i.e., that $\langle g \cdot u, w \rangle = 0$ for all $w \in W$.

Since W is a G -submodule, $g^{-1} \cdot w \in W$ for all $w \in W$. Using G -invariance of the inner product:

$$\langle g \cdot u, w \rangle = \langle g^{-1} \cdot (g \cdot u), g^{-1} \cdot w \rangle \quad (2.9)$$

$$= \langle u, g^{-1} \cdot w \rangle \quad (2.10)$$

$$= 0, \quad (2.11)$$

where the last equality holds because $u \in W^\perp$ and $g^{-1} \cdot w \in W$.

Therefore $g \cdot u \in W^\perp$.

Since $W^\perp$ is a G -submodule and $W \cap W^\perp = \{0\}$, we have

$$V = W \oplus W^\perp, \quad (2.12)$$

a decomposition as G -modules. $\square$

Theorem 2.3 (Theorem L4-3.2 Maschke's Theorem - ONLY READ THROUGH (WILL NOT BE ON EXAM)). *Let G be a finite group and V a finitely generated $\mathbb{C}G$ -module (i.e., a finite-dimensional representation of G over $\mathbb{C}$). Then:*

$$(i) \quad \text{Every } G\text{-submodule of } V \text{ has a } G\text{-complement.} \quad (2.13)$$

$$(ii) \quad V = W^{(1)} \oplus_G W^{(2)} \oplus_G \cdots \oplus_G W^{(k)}, \quad (2.14)$$

where each $W^{(i)}$ is an irreducible G -module.

Proof. Before proving the theorem, we show that finiteness of G is essential.

Example. Let $G = (\mathbb{R}_{>0}, \cdot, 1)$ and define $\rho : G \rightarrow \text{GL}_2(\mathbb{C})$ by

$$\rho(r) = \begin{pmatrix} 1 & \log r \\ 0 & 1 \end{pmatrix}. \quad (2.15)$$

For $r, s \in G$,

$$\rho(r)\rho(s) = \begin{pmatrix} 1 & \log r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \log s \\ 0 & 1 \end{pmatrix} \quad (2.16)$$

$$= \begin{pmatrix} 1 & \log r + \log s \\ 0 & 1 \end{pmatrix} \quad (2.17)$$

$$= \begin{pmatrix} 1 & \log(rs) \\ 0 & 1 \end{pmatrix} = \rho(rs). \quad (2.18)$$

Let $W = \mathbb{C}\{e_1\}$. Then W is G -invariant since $\rho(r)e_1 = e_1$.

Suppose $U = \mathbb{C} \begin{pmatrix} a \\ b \end{pmatrix}$ is a G -complement with $b \neq 0$. Then invariance requires

$$\rho(r) \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a + b \log r \\ b \end{pmatrix} \in U, \quad (2.19)$$

forcing

$$\frac{a + b \log r}{b} = \frac{a}{b}, \quad (2.20)$$

so $\log r = 0$ for all r , a contradiction.

(i) Let $W \subseteq V$ be a G -submodule. Choose a basis $\{v_1, \dots, v_n\}$ of V and define

$$\langle v_i, v_j \rangle = \delta_{ij}. \quad (2.21)$$

Define a new form

$$\langle u, v \rangle' = \sum_{g \in G} \langle g \cdot u, g \cdot v \rangle. \quad (2.22)$$

This is a Hermitian inner product: linearity and conjugate symmetry follow termwise, and

$$\langle u, u \rangle' = \sum_{g \in G} \langle g \cdot u, g \cdot u \rangle \geq 0, \quad (2.23)$$

with equality only if $u = 0$.

It is G -invariant since for $h \in G$,

$$\langle h \cdot u, h \cdot v \rangle' = \sum_{g \in G} \langle gh \cdot u, gh \cdot v \rangle \quad (2.24)$$

$$= \sum_{g' \in G} \langle g' \cdot u, g' \cdot v \rangle \quad (2.25)$$

$$= \langle u, v \rangle'. \quad (2.26)$$

Thus $W^\perp$ is a G -submodule and

$$V = W \oplus W^\perp. \quad (2.27)$$

(ii) We proceed by induction on $\dim V$.

If $\dim V = 1$, then V is irreducible.

If V is reducible, there exists $0 \subsetneq W \subsetneq V$. By (i),

$$V = W \oplus W'. \quad (2.28)$$

Since $\dim W, \dim W' < \dim V$, both decompose into irreducibles. Combining gives

$$V = W^{(1)} \oplus_G \cdots \oplus_G W^{(k)}. \quad (2.29)$$

□

Proposition 2.4 (Proposition L5-1.1: One-dimensional representations of $\mathbb{Z}_n$). *Let $\mathbb{Z}_n = \langle g \rangle$ be the cyclic group of order n and let $\rho : \mathbb{Z}_n \rightarrow \text{GL}_1(\mathbb{C})$ be a representation. Then*

$$\rho(g) = e^{2\pi i q/n} \quad (2.30)$$

for some $q \in \{0, 1, \dots, n-1\}$.

In particular, there are exactly n one-dimensional representations of $\mathbb{Z}_n$, given by

$$\rho_q(g) = e^{2\pi i q/n}, \quad q = 0, 1, \dots, n-1. \quad (2.31)$$

Proof. Since $g^n = e$ and ρ is a homomorphism, we have

$$\rho(g)^n = \rho(g^n) = \rho(e) = 1. \quad (2.32)$$

So $\rho(g)$ is an n -th root of unity in $\mathbb{C}^\times$. The n -th roots of unity are precisely

$$e^{2\pi i q/n}, \quad q = 0, 1, \dots, n-1. \quad (2.33)$$

Each choice of q determines a unique homomorphism $\rho_q : \mathbb{Z}_n \rightarrow \text{GL}_1(\mathbb{C})$ via

$$\rho_q(g^k) = e^{2\pi i qk/n}. \quad (2.34)$$

One easily verifies that ρ_q is indeed a group homomorphism. □

Proposition 2.5 (Proposition L5-3.2: Isomorphic modules have the same character). *If $V \cong W$ as G -modules, then*

$$\chi^V = \chi^W. \quad (2.35)$$

Proof. Let $\phi : V \rightarrow W$ be a G -module isomorphism. Then there exist bases B_V of V and B_W of W such that

$$M_{B_V}(g) = M_{B_W}(g) \quad (2.36)$$

for all $g \in G$.

Indeed, if $B_V = \{v_1, \dots, v_n\}$, take

$$B_W = \{\phi(v_1), \dots, \phi(v_n)\}. \quad (2.37)$$

Therefore, for all $g \in G$,

$$\chi_V(g) = \text{Tr}(M_{B_V}(g)) = \text{Tr}(M_{B_W}(g)) = \chi_W(g). \quad (2.38)$$

□

Remark 2.1 (Observation). Characters are **not** group homomorphisms in general. That is, $\chi^V(gh) \neq \chi^V(g) \cdot \chi^V(h)$ in general, because $\text{Tr}(AB) \neq \text{Tr}(A)\text{Tr}(B)$ in general.

Proposition 2.6 (Proposition L5-4.1: Fundamental properties of characters (ONLY NEED TO KNOW (i))). *Let V be a G -module of dimension n with character $\chi = \chi_V$. Then:*

$$(i) \quad \chi(g^{-1}) = \chi(g) \quad \text{for all } g \in G, \quad (2.39)$$

$$(ii) \quad |\chi(g)| \leq n \quad \text{for all } g \in G, \quad (2.40)$$

with equality in (ii) if and only if

$$\rho(g) = \lambda \cdot \text{id}_V \quad (2.41)$$

for some $\lambda \in \mathbb{C}$.

Proof. Let $g \in G$ and let $k = |g|$. Since $\langle g \rangle$ is cyclic (hence abelian), the restriction of ρ to $\langle g \rangle$ can be simultaneously diagonalized: there exists a basis B of V such that

$$M_B(g) = \text{diag}(\lambda_1, \dots, \lambda_n). \quad (2.42)$$

Since $g^k = e$, each eigenvalue satisfies

$$\lambda_i^k = 1, \quad (2.43)$$

so

$$\lambda_i = e^{2\pi i q_i / k} \quad (2.44)$$

for some integer q_i . In particular,

$$|\lambda_i| = 1, \quad \lambda_i^{-1} = \overline{\lambda_i}. \quad (2.45)$$

(i) We have

$$M_B(g^{-1}) = M_B(g)^{-1} = \text{diag}(\lambda_1^{-1}, \dots, \lambda_n^{-1}), \quad (2.46)$$

so

$$\chi(g^{-1}) = \sum_{i=1}^n \lambda_i^{-1} \quad (2.47)$$

$$= \sum_{i=1}^n \overline{\lambda_i} \quad (2.48)$$

$$= \overline{\sum_{i=1}^n \lambda_i} \quad (2.49)$$

$$= \overline{\chi(g)}. \quad (2.50)$$

Since $\chi(g)$ is a sum of roots of unity, it follows that $\chi(g^{-1}) = \chi(g)$.

(ii) By the triangle inequality,

$$|\chi(g)| = \left| \sum_{i=1}^n \lambda_i \right| \leq \sum_{i=1}^n |\lambda_i| = n. \quad (2.51)$$

Equality holds if and only if all λ_i have the same argument, i.e.,

$$\lambda_1 = \cdots = \lambda_n = \lambda. \quad (2.52)$$

In this case,

$$M_B(g) = \lambda I_n, \quad (2.53)$$

so

$$\rho(g) = \lambda \cdot \text{id}_V. \quad (2.54)$$

□

Corollary 2.7 (Corollary L6-2.3: Irreducibility criterion). *A G -module V is irreducible if and only if*

$$\langle \chi_V, \chi_V \rangle = 1. \quad (2.55)$$

Proof. Write

$$V \cong \bigoplus_i n_i V_i, \quad (2.56)$$

where the V_i are pairwise non-isomorphic irreducible G -modules.

By row orthogonality,

$$\langle \chi_V, \chi_V \rangle = \sum_i n_i^2. \quad (2.57)$$

This equals 1 if and only if exactly one $n_i = 1$ and all others are 0, which is precisely the condition that V is irreducible. □

Theorem 2.8 (Theorem L6-2.6: Characters determine representations). *Let V and W be G -modules.*

Then

$$V \cong_G W \iff \chi_V = \chi_W. \quad (2.58)$$

Proof. The forward direction ($V \cong_G W \Rightarrow \chi_V = \chi_W$) was established in Proposition 2.

For the converse, suppose $\chi_V = \chi_W$. Let $\{V_1, \dots, V_r\}$ be a complete set of pairwise non-isomorphic irreducible G -modules. By Maschke's Theorem, we can decompose

$$V \cong \bigoplus_{i=1}^r n_i V_i, \quad W \cong \bigoplus_{i=1}^r m_i V_i, \quad (2.59)$$

where n_i and m_i are the multiplicities.

By the multiplicity formula,

$$n_i = \langle \chi_{V_i}, \chi_V \rangle = \langle \chi_{V_i}, \chi_W \rangle = m_i \quad (2.60)$$

for all i .

Since V and W have the same irreducible decomposition, it follows that

$$V \cong_G W. \quad (2.61)$$

□

Corollary 2.9 (Corollary L6-2.7: Irreducible characters are linearly independent). *The irreducible characters $\chi_{V_1}, \dots, \chi_{V_r}$ form an orthonormal set in F_G (with respect to $\langle \cdot, \cdot \rangle$). In particular,*

$$\{\chi_{V_1}, \dots, \chi_{V_r}\} \text{ are linearly independent.} \quad (2.62)$$

Proof. Orthonormality is exactly the content of Theorem 1. Any orthonormal set in an inner product space is linearly independent: if

$$\sum_i c_i \chi_{V_i} = 0, \quad (2.63)$$

then taking the inner product with χ_{V_j} gives

$$c_j = \left\langle \chi_{V_j}, \sum_i c_i \chi_{V_i} \right\rangle = 0. \quad (2.64)$$

□

Theorem 2.10 (Theorem L6-5.1: Decomposition of the regular representation). *Let G be a finite group and $\mathbb{C}G$ its regular representation. For each irreducible G -module V , the multiplicity of V in $\mathbb{C}G$ is $\dim V$. Thus*

$$\mathbb{C}G \cong_G (\dim V_1) V_1 \oplus (\dim V_2) V_2 \oplus \cdots \oplus (\dim V_r) V_r, \quad (2.65)$$

where $V_1, V_2, \dots, V_r$ are the distinct irreducible G -modules.

Proof. Recall the character of the regular representation:

$$\chi_{\mathbb{C}G}(g) = \begin{cases} |G| & \text{if } g = e, \\ 0 & \text{if } g \neq e. \end{cases} \quad (2.66)$$

Let V be an irreducible G -module. By the multiplicity formula,

$$\text{mult}(V \text{ in } \mathbb{C}G) = \langle \chi_V, \chi_{\mathbb{C}G} \rangle \quad (2.67)$$

$$= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathbb{C}G}(g) \overline{\chi_V(g)} \quad (2.68)$$

$$= \frac{1}{|G|} \cdot |G| \cdot \overline{\chi_V(e)} \quad (2.69)$$

$$= \overline{\chi_V(e)} \quad (2.70)$$

$$= \chi_V(e) \quad (2.71)$$

$$= \dim V. \quad (2.72)$$

Here we used that $\chi_V(e) = \dim V$, which is a positive real number. $\square$

2.2 Important Definitions for The Final Exam

Note 2. Here is an outline of the totality of important definitions from the representation theory lectures:

1. Lecture 1: Foundations

- (a) **Definition:** Linear action
- (b) **Definition:** Linear representation
- (c) **Definition:** Matrix representation
- (d) **Proposition:** Equivalence of actions, representations, and matrix representations
- (e) **Definition:** Isomorphism of linear representations
- (f) **Definition:** Isomorphism of G -sets
- (g) **Definition:** Isomorphism of matrix representations
- (h) **Example:** Trivial representation
- (i) **Example:** Sign representation of S_n
- (j) **Example:** Natural (defining) representation
- (k) **Example:** Permutation representation
- (l) **Example:** Regular representation

2. Lecture 2: Modules and Group Algebra

- (a) **Definition:** R -module
- (b) **Definition:** Submodule
- (c) **Definition:** Simple (irreducible) module
- (d) **Definition:** Finitely generated module
- (e) **Definition:** Direct sum (internal and external)
- (f) **Definition:** Direct summand
- (g) **Definition:** Module homomorphism
- (h) **Theorem:** First Isomorphism Theorem (modules)
- (i) **Definition:** Group algebra RG
- (j) **Definition:** R -algebra
- (k) **Definition:** Finite-dimensional algebra

- (l) **Theorem:** Four-way equivalence (modules $\leftrightarrow$ representations)

3. Lecture 3: Reducibility and Schur

- (a) **Definition:** G -submodule
- (b) **Definition:** Irreducible representation
- (c) **Proposition:** Checking G -invariance
- (d) **Theorem:** Irreducible representations of abelian groups are 1-dimensional
- (e) **Corollary:** Simultaneous diagonalization for abelian groups
- (f) **Corollary:** Every element acts diagonalizably

4. Lecture 4: Maschke and Structure

- (a) **Definition:** G -complement
- (b) **Definition:** Hermitian inner product
- (c) **Definition:** Orthogonal complement
- (d) **Proposition:** Orthogonal decomposition
- (e) **Definition:** G -invariant inner product
- (f) **Proposition:** Orthogonal complements of G -submodules
- (g) **Definition:** Semisimple module
- (h) **Corollary:** Block diagonalization of representations
- (i) **Corollary:** Relationship between complements and quotients
- (j) **Proposition:** Uniqueness of complements up to isomorphism

5. Lecture 5: Characters

- (a) **Definition:** Trace
- (b) **Proposition:** Trace is cyclic
- (c) **Proposition:** Trace is conjugation-invariant
- (d) **Definition:** Character of a representation
- (e) **Proposition:** Isomorphic representations have same character
- (f) **Proposition:** Basic properties of characters
- (g) **Proposition:** Character of a direct sum
- (h) **Proposition:** One-dimensional representations of $\mathbb{Z}_n$

6. Lecture 6: Orthogonality (Conceptual Only)

- (a) **Definition:** Inner product on class functions
- (b) **Definition:** Class function
- (c) **Concept:** Orthogonality of characters (rows/columns)
- (d) **Concept:** Number of irreducibles equals number of conjugacy classes
- (e) **Concept:** Regular representation and dimension formula

2.2.1 Lecture 1: Foundations

Definition 2.1 (Linear action). Let k be a field, G a group, and V a k -vector space. An action $\alpha : G \times V \rightarrow V$ is called a **linear action** if for every $g \in G$, the map $v \mapsto g \cdot v$ is k -linear. That is:

$$g \cdot (x + y) = g \cdot x + g \cdot y, \quad (2.73)$$

$$g \cdot (\lambda x) = \lambda(g \cdot x), \quad (2.74)$$

for all $x, y \in V$, $\lambda \in k$, and $g \in G$.

Definition 2.2 (Linear representation). A **linear representation** of a group G on a k -vector space V is a group homomorphism

$$\rho : G \rightarrow \text{GL}(V), \quad (2.75)$$

where $\text{GL}(V)$ is the group of invertible k -linear maps $V \rightarrow V$.

Definition 2.3 (Matrix representation). A **matrix representation** of G of degree n over k is a group homomorphism

$$\rho : G \rightarrow \text{GL}_n(k). \quad (2.76)$$

Proposition 2.11 (Equivalence of actions, representations, and matrix representations). *Let V be a k -vector space with $\dim V = n$ and G a group. The following are equivalent:*

$$(i) \quad \text{Linear actions of } G \text{ on } V, \quad (2.77)$$

$$(ii) \quad \text{Linear representations } \rho : G \rightarrow \text{GL}(V), \quad (2.78)$$

$$(iii) \quad \text{Matrix representations } \rho : G \rightarrow \text{GL}_n(k). \quad (2.79)$$

Definition 2.4 (Isomorphism of linear representations). Let $\rho_1 : G \rightarrow \text{GL}(V)$ and $\rho_2 : G \rightarrow \text{GL}(W)$ be representations. We say they are **isomorphic** (write $V \cong_G W$) if there exists a linear isomorphism $\phi : V \rightarrow W$ such that

$$\rho_2(g) \circ \phi = \phi \circ \rho_1(g) \quad (2.80)$$

for all $g \in G$.

Definition 2.5 (Isomorphism of G -sets). Let X and Y be G -sets. A **G -set isomorphism** is a bijection $\phi : X \rightarrow Y$ such that

$$\phi(g \cdot x) = g \cdot \phi(x) \quad (2.81)$$

for all $g \in G$ and $x \in X$.

Definition 2.6 (Isomorphism of matrix representations). Two matrix representations $\rho_1, \rho_2 : G \rightarrow \text{GL}_n(\mathbb{C})$ are **isomorphic** if there exists $A \in \text{GL}_n(\mathbb{C})$ such that

$$\rho_2(g) = A \rho_1(g) A^{-1} \quad (2.82)$$

for all $g \in G$.

Example 2.2 (Trivial representation). Let G be a group and $V = \mathbb{C}$. The trivial representation is

$$\mathbf{1}_G : G \rightarrow \text{GL}(\mathbb{C}), \quad g \mapsto \text{id}_{\mathbb{C}}. \quad (2.83)$$

Equivalently,

$$g \cdot x = x \quad \text{for all } g \in G, x \in \mathbb{C}. \quad (2.84)$$

Example 2.3 (Sign representation of S_n). Let $G = S_n$ and $V = \mathbb{C}$. The sign representation is

$$\text{sgn} : S_n \rightarrow \mathbb{C}^\times, \quad \sigma \mapsto \text{sgn}(\sigma), \quad (2.85)$$

where

$$\text{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even,} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases} \quad (2.86)$$

Example 2.4 (Natural (defining) representation). Let $G = S_n$ and $V = \mathbb{C}[n]$ with basis $\{1, 2, \dots, n\}$. Define

$$\sigma \cdot i = \sigma(i), \quad (2.87)$$

and extend linearly:

$$\sigma \cdot \left(\sum_{i=1}^n \alpha_i i \right) = \sum_{i=1}^n \alpha_i \sigma(i). \quad (2.88)$$

This defines a representation of degree n .

Example 2.5 (Permutation representation). Let G act on a finite set $X = \{x_1, \dots, x_n\}$. Then the associated permutation representation is

$$\rho : G \rightarrow \text{GL}_n(\mathbb{C}), \quad (2.89)$$

where $V = \mathbb{C}X$ and

$$g \cdot \left(\sum_{x \in X} \alpha_x x \right) = \sum_{x \in X} \alpha_x (g \cdot x). \quad (2.90)$$

Example 2.6 (Regular representation). Let G be a finite group. The regular representation is the action of G on itself by left multiplication:

$$g \cdot x = gx. \quad (2.91)$$

The representation space is

$$\mathbb{C}G = \left\{ \sum_{g \in G} a_g g \mid a_g \in \mathbb{C} \right\}, \quad (2.92)$$

and the representation is given by

$$\rho(g) \left(\sum_{h \in G} a_h h \right) = \sum_{h \in G} a_h (gh). \quad (2.93)$$

2.2.2 Lecture 2: Modules and Group Algebra

Definition 2.7 (R -module). Let R be a ring with unity. A (left) R -module is an abelian group $(M, +)$ together with a map

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm, \quad (2.94)$$

satisfying:

$$1 \cdot m = m, \quad (2.95)$$

$$r(m + n) = rm + rn, \quad (2.96)$$

$$(r + s)m = rm + sm, \quad (2.97)$$

$$r(sm) = (rs)m, \quad (2.98)$$

for all $r, s \in R$ and $m, n \in M$.

Definition 2.8 (Submodule). Let M be an R -module. A subset $N \subseteq M$ is a submodule if:

$$N \leq (M, +) \quad \text{and} \quad rn \in N \text{ for all } r \in R, n \in N. \quad (2.99)$$

Definition 2.9 (Simple (irreducible) module). An R -module $M \neq \{0\}$ is called simple (or irreducible) if its only submodules are

$$\{0\} \quad \text{and} \quad M. \quad (2.100)$$

Definition 2.10 (Finitely generated module). An R -module M is finitely generated if there exist $m_1, \dots, m_k \in M$ such that

$$M = \left\{ \sum_{i=1}^k a_i m_i \mid a_i \in R \right\}. \quad (2.101)$$

Definition 2.11 (Direct sum (internal and external)). Let N_1, N_2 be submodules of M .

$$N_1 + N_2 = \{n_1 + n_2 \mid n_1 \in N_1, n_2 \in N_2\}. \quad (2.102)$$

If additionally

$$N_1 \cap N_2 = \{0\}, \quad (2.103)$$

then we write

$$M = N_1 \oplus N_2. \quad (2.104)$$

For R -modules M and N , the external direct sum is

$$M \oplus N = M \times N, \quad (2.105)$$

with componentwise operations.

Definition 2.12 (Direct summand). A submodule $N \subseteq M$ is a direct summand if there exists a submodule N' such that

$$M = N \oplus N'. \quad (2.106)$$

Definition 2.13 (Module homomorphism). Let M, N be R -modules. A function $f : M \rightarrow N$ is an R -module homomorphism if

$$f(m_1 + m_2) = f(m_1) + f(m_2), \quad (2.107)$$

$$f(rm) = rf(m), \quad (2.108)$$

for all $r \in R$ and $m, m_1, m_2 \in M$.

Theorem 2.12 (First Isomorphism Theorem for modules). *Let $f : M \rightarrow N$ be an R -module homomorphism. Then:*

$$\ker f \leq M, \quad (2.109)$$

$$\operatorname{im} f \leq N, \quad (2.110)$$

$$M / \ker f \cong \operatorname{im} f. \quad (2.111)$$

Definition 2.14 (Group algebra RG). Let G be a finite group and R a commutative ring. The group algebra is

$$RG = \left\{ \sum_{g \in G} a_g g \mid a_g \in R \right\}, \quad (2.112)$$

with multiplication defined by

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{h \in G} b_h h \right) = \sum_{g, h \in G} a_g b_h (gh). \quad (2.113)$$

Definition 2.15 (R -algebra). An R -algebra is an R -module A equipped with a bilinear multiplication

$$A \times A \rightarrow A \quad (2.114)$$

such that

$$(\lambda a)b = \lambda(ab) = a(\lambda b), \quad (2.115)$$

for all $\lambda \in R$ and $a, b \in A$.

Definition 2.16 (Finite-dimensional algebra). A k -algebra A (where k is a field) is finite-dimensional if

$$\dim_k A < \infty. \quad (2.116)$$

Theorem 2.13 (Four-way equivalence (modules $\leftrightarrow$ representations)). *Let G be a finite group. There is a bijective correspondence between:*

$$(i) \text{ Finitely generated } \mathbb{C}G\text{-modules,} \quad (2.117)$$

$$(ii) \text{ Linear actions of } G \text{ on finite-dimensional vector spaces,} \quad (2.118)$$

$$(iii) \text{ Linear representations } \rho : G \rightarrow \text{GL}(V), \quad (2.119)$$

$$(iv) \text{ Matrix representations } \rho : G \rightarrow \text{GL}_n(\mathbb{C}). \quad (2.120)$$

2.2.3 Lecture 3: Reducibility and Schur

Definition 2.17 (G -submodule). Let V be a G -module. A subspace $W \subseteq V$ is called a G -submodule (or G -invariant subspace) if

$$g \cdot w \in W \quad (2.121)$$

for all $g \in G$ and $w \in W$.

Definition 2.18 (Irreducible representation). A G -module $V \neq \{0\}$ is called irreducible (or simple) if its only G -submodules are

$$\{0\} \quad \text{and} \quad V. \quad (2.122)$$

Proposition 2.14 (Checking G -invariance). *Let V be a G -module and $W \subseteq V$ a subspace with basis $B_W = \{w_1, \dots, w_k\}$. Then W is a G -submodule if and only if*

$$g \cdot w_i \in W \quad (2.123)$$

for all $g \in G$ and all $w_i \in B_W$.

Moreover, it suffices to check this for a generating set $\{s_1, \dots, s_m\}$ of G , i.e.,

$$s_j \cdot w_i \in W \quad \forall i, j \implies W \text{ is a } G\text{-submodule.} \quad (2.124)$$

Theorem 2.15 (Irreducible representations of abelian groups are one-dimensional). *Let G be a finite abelian group and V an irreducible representation of G over $\mathbb{C}$. Then*

$$\dim V = 1. \quad (2.125)$$

Corollary 2.16 (Simultaneous diagonalization for abelian groups). *Let G be a finite abelian group and $\rho : G \rightarrow \text{GL}(V)$ a representation. Then there exists a basis B of V such that*

$$M_B(g) \text{ is diagonal for all } g \in G. \quad (2.126)$$

Corollary 2.17 (Every element acts diagonalizably). *Let G be a finite group, $\rho : G \rightarrow \text{GL}(V)$ a representation, and $g \in G$. Then there exists a basis of V such that*

$$M(g) \text{ is diagonal.} \quad (2.127)$$

In particular, every $\rho(g)$ is diagonalizable.

2.2.4 Lecture 4: Maschke and Structure

Definition 2.19 (G -complement). Let V be a G -module and $W \subseteq V$ a G -submodule. A submodule $W' \subseteq V$ is called a G -complement of W if

$$V = W \oplus_G W', \quad (2.128)$$

i.e.,

$$V = W + W' \quad \text{and} \quad W \cap W' = \{0\}. \quad (2.129)$$

Definition 2.20 (Hermitian inner product). A Hermitian inner product on a $\mathbb{C}$ -vector space V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C} \quad (2.130)$$

such that for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{C}$:

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle, \quad (2.131)$$

$$\langle v, w \rangle = \overline{\langle w, v \rangle}, \quad (2.132)$$

$$\langle v, v \rangle \geq 0, \quad \langle v, v \rangle = 0 \iff v = 0. \quad (2.133)$$

Definition 2.21 (Orthogonal complement). Let V be a vector space with Hermitian inner product and $W \subseteq V$ a subspace. The orthogonal complement of W is

$$W^\perp = \{v \in V \mid \langle v, w \rangle = 0 \text{ for all } w \in W\}. \quad (2.134)$$

Proposition 2.18 (Orthogonal decomposition). *Let V be a finite-dimensional $\mathbb{C}$ -vector space with Hermitian inner product and $W \subseteq V$ a subspace. Then*

$$V = W \oplus W^\perp. \quad (2.135)$$

Definition 2.22 (G -invariant inner product). Let V be a G -module. A Hermitian inner product $\langle \cdot, \cdot \rangle$ is called G -invariant if

$$\langle g \cdot u, g \cdot v \rangle = \langle u, v \rangle \quad (2.136)$$

for all $g \in G$ and $u, v \in V$.

Proposition 2.19 (Orthogonal complements of G -submodules). *Let V be a G -module with a G -invariant inner product and let $W \subseteq V$ be a G -submodule. Then*

$$W^\perp \text{ is a } G\text{-submodule, and } V = W \oplus_G W^\perp. \quad (2.137)$$

Definition 2.23 (Semisimple module). A G -module V is called semisimple (or completely reducible) if

$$V = W^{(1)} \oplus_G W^{(2)} \oplus_G \cdots \oplus_G W^{(k)}, \quad (2.138)$$

where each $W^{(i)}$ is an irreducible G -submodule.

Corollary 2.20 (Block diagonalization of representations). *Let $\rho : G \rightarrow \text{GL}_n(\mathbb{C})$ be a representation of a finite group. Then there exists $A \in \text{GL}_n(\mathbb{C})$ such that*

$$\rho(g) = A \begin{pmatrix} \rho_1(g) & & \\ & \ddots & \\ & & \rho_k(g) \end{pmatrix} A^{-1} \quad (2.139)$$

for all $g \in G$, where each ρ_i is irreducible.

Corollary 2.21 (Relationship between complements and quotients). *Let V be a G -module and $W \subseteq V$ a G -submodule. Then:*

$$(i) \quad V/W \text{ is a } G\text{-module}, \quad (2.140)$$

$$(ii) \quad W' \cong_G V/W \text{ for any } G\text{-complement } W', \quad (2.141)$$

$$(iii) \quad V \cong_G W \oplus_G V/W. \quad (2.142)$$

Proposition 2.22 (Uniqueness of complements up to isomorphism). *Let V be a G -module and $W \subseteq V$ a G -submodule. If W_1 and W_2 are G -complements of W , then*

$$W_1 \cong_G W_2. \quad (2.143)$$

2.2.5 Lecture 5: Characters

Definition 2.24 (Trace). The trace is the function

$$\text{Tr} : M_n(\mathbb{C}) \rightarrow \mathbb{C} \quad (2.144)$$

defined by

$$\text{Tr}(A) = \sum_{i=1}^n A_{ii}, \quad (2.145)$$

i.e., the sum of the diagonal entries of A .

Proposition 2.23 (Trace is cyclic). *For all $A, B \in M_n(\mathbb{C})$,*

$$\text{Tr}(AB) = \text{Tr}(BA). \quad (2.146)$$

Proposition 2.24 (Trace is conjugation-invariant). *For all $A \in M_n(\mathbb{C})$ and invertible $P \in \text{GL}_n(\mathbb{C})$,*

$$\text{Tr}(PAP^{-1}) = \text{Tr}(A). \quad (2.147)$$

Definition 2.25 (Character of a representation). Let V be a G -module with representation $\rho : G \rightarrow \text{GL}(V)$. The character of V is the function

$$\chi_V : G \rightarrow \mathbb{C}, \quad \chi_V(g) = \text{Tr}(M_B(g)), \quad (2.148)$$

where $M_B(g)$ is the matrix of $\rho(g)$ with respect to any basis B of V .

Proposition 2.25 (Isomorphic representations have the same character). *If $V \cong_G W$, then*

$$\chi_V = \chi_W. \quad (2.149)$$

Proposition 2.26 (Basic properties of characters). *Let V be a G -module of dimension n with character $\chi = \chi_V$. Then:*

$$(i) \quad \chi(g^{-1}) = \overline{\chi(g)}, \quad (2.150)$$

$$(ii) \quad |\chi(g)| \leq n, \quad (2.151)$$

with equality in (ii) if and only if

$$\rho(g) = \lambda \cdot \text{id}_V \quad (2.152)$$

for some $\lambda \in \mathbb{C}$.

Proposition 2.27 (Character of a direct sum). *If $V = W \oplus U$ as G -modules, then*

$$\chi_V = \chi_W + \chi_U. \quad (2.153)$$

Proposition 2.28 (One-dimensional representations of $\mathbb{Z}_n$). *Let $\mathbb{Z}_n = \langle g \rangle$ and $\rho : \mathbb{Z}_n \rightarrow \text{GL}_1(\mathbb{C})$ a representation. Then*

$$\rho(g) = e^{2\pi i q/n} \quad (2.154)$$

for some $q \in \{0, 1, \dots, n-1\}$.

In particular,

$$\rho_q(g) = e^{2\pi i q/n}, \quad q = 0, 1, \dots, n-1, \quad (2.155)$$

are all the one-dimensional representations.

2.2.6 Lecture 6: Orthogonality (Conceptual Only)

Definition 2.26 (Inner product on class functions). Let G be a finite group and let F_G denote the space of complex-valued functions on G . The inner product on class functions is defined by

$$\langle f, g \rangle = \frac{1}{|G|} \sum_{x \in G} f(x) \overline{g(x)}. \quad (2.156)$$

Definition 2.27 (Class function). A function $f : G \rightarrow \mathbb{C}$ is called a class function if

$$f(hgh^{-1}) = f(g) \quad (2.157)$$

for all $g, h \in G$, i.e., f is constant on conjugacy classes.

Proposition 2.29 (Orthogonality of characters (conceptual)). *Let $\{\chi_1, \dots, \chi_r\}$ be the irreducible characters of G . Then:*

$$\langle \chi_i, \chi_j \rangle = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases} \quad (2.158)$$

That is, irreducible characters form an orthonormal set (row orthogonality).

There is also a dual statement (column orthogonality) relating values across conjugacy classes.

Theorem 2.30 (Number of irreducibles equals number of conjugacy classes). *The number of distinct irreducible representations of G (up to isomorphism) is equal to the number of conjugacy classes of G .*

Theorem 2.31 (Regular representation and dimension formula). *Let G be a finite group with irreducible representations $V_1, \dots, V_r$. Then:*

$$\mathbb{C}G \cong_G (\dim V_1)V_1 \oplus \dots \oplus (\dim V_r)V_r, \quad (2.159)$$

and consequently,

$$|G| = \sum_{i=1}^r (\dim V_i)^2. \quad (2.160)$$

2.3 Will's Answers for Homework 3

Result 2.1 (Problem 1). We can observe the elements in $\mathfrak{S}_3 = \{e, (12), (13), (23), (123), (132)\}$. Now we can create matrix representations for the permutation of elements in the set $\{1, 2, 3\}$

$$\begin{aligned} M(e) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & M((12)) &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} & M((13)) &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ M((23)) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} & M((123)) &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} & M((132)) &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \end{aligned}$$

Problem 2. Assume for the sake of contradiction that the sign representation $\text{sgn} : \mathfrak{S}_n \rightarrow \mathbb{C}^*$ is a permutation representation. This means that there exists a $\mathfrak{S}_n$ -isomorphism, φ , where $\varphi : \mathbb{C} \rightarrow \mathbb{C}X$. We also say that $\exists X = \{x_1, x_2, \dots, x_n\}$ with $|X| = n \geq 2$ making this true. We let $|X| \geq 2$ since if $|X| = 1$ the permutation representation would turn into the trivial representation. The representations are described as follows:

$$\rho : \mathfrak{S}_n \rightarrow \text{GL}_n(\mathbb{C}) \quad (2.161)$$

is the permutation representation, and

$$\text{sgn} : \mathfrak{S}_n \rightarrow \text{GL}_1(\mathbb{C}) \cong \mathbb{C}^* \quad (2.162)$$

is the sign representation. We also say that the vector space associated with ρ is $\mathbb{C}X$ and the vector space associated with sgn is $\mathbb{C}$. Since we have assumed that φ is a $\mathfrak{S}_n$ -isomorphism, we know that we must have $\rho(\sigma) \circ \varphi = \varphi \circ \text{sgn}(\sigma)$ for all $\sigma \in \mathfrak{S}_n$. Yet, we can see that this is impossible. We know that $\dim(\mathbb{C}) = 1$ and we know that $\mathbb{C}X \cong \mathbb{C}^n$ and since $\dim(\mathbb{C}^n) = n \geq 2$, we know that $\dim(\mathbb{C}X) = n$ since $\dim(\mathbb{C}) \neq \dim \mathbb{C}X$. This means that it is in fact impossible for φ to be an isomorphism between $\mathbb{C}$ and $\mathbb{C}X$. Here we arrive at a contradiction. So it must be the case that the sign representation is not a permutation representation. $\square$

Problem 3. We can see through some calculations that $H_1 = \{e, (12)\}$, $H_2 = \{e, (13)\}$, $H_3 = \{e, (23)\}$. From this we get that

$$\mathfrak{S}_3/H_1 = \{eH_1, (13)H_1, (23)H_1\} \quad (2.163)$$

$$\mathfrak{S}_3/H_2 = \{eH_2, (12)H_2, (23)H_2\} \quad (2.164)$$

$$\mathfrak{S}_3/H_3 = \{eH_3, (12)H_3, (13)H_3\}. \quad (2.165)$$

The representations of each of these cosets can be described as follows:

$$\rho_1 : \mathfrak{S}_3 \rightarrow \text{GL}(\mathfrak{S}_3/H_1), \quad (2.166)$$

$$\rho_2 : \mathfrak{S}_3 \rightarrow \text{GL}(\mathfrak{S}_3/H_2), \quad (2.167)$$

$$\rho_3 : \mathfrak{S}_3 \rightarrow \text{GL}(\mathfrak{S}_3/H_3). \quad (2.168)$$

In order to show that each of these representations are isomorphic we must construct $\mathfrak{S}_3$ -isomorphisms τ_1, τ_2 such that $\rho_2(\sigma) \circ \tau_1 = \tau_1 \circ \rho_1(\sigma)$, and $\rho_3(\sigma) \circ \tau_2 = \tau_2 \circ \rho_2(\sigma)$. We also require τ_1, τ_2 are linear isomorphisms. We will define τ_1 and τ_2 as such:

$$\tau_1 : \mathbb{C}\{\mathfrak{S}_3/H_1\} \rightarrow \mathbb{C}\{\mathfrak{S}_3/H_2\} \quad (2.169)$$

$$\tau_2 : \mathbb{C}\{\mathfrak{S}_3/H_2\} \rightarrow \mathbb{C}\{\mathfrak{S}_3/H_3\}. \quad (2.170)$$

Definition of τ_1 . Define

$$\tau_1 : \mathbb{C}\{\mathfrak{S}_3/H_1\} \rightarrow \mathbb{C}\{\mathfrak{S}_3/H_2\} \quad (2.171)$$

by

$$\tau_1(\sigma H_1) = \sigma(23)H_2. \quad (2.172)$$

Then for $aeH_1 + b(13)H_1 + c(23)H_1 \in \mathbb{C}\{\mathfrak{S}_3/H_1\}$, we have

$$\tau_1(aeH_1 + b(13)H_1 + c(23)H_1) = a\tau_1(eH_1) + b\tau_1((13)H_1) + c\tau_1((23)H_1) \quad (2.173)$$

$$= a(23)H_2 + b(13)(23)H_2 + c(23)(23)H_2 \quad (2.174)$$

$$= a(23)H_2 + b(132)H_2 + ceH_2. \quad (2.175)$$

$$= a(23)H_2 + b(12)H_2 + ceH_2 \in \mathfrak{S}_3/H_2. \quad (2.176)$$

Inverse of τ_1 . Define

$$\tau_1^{-1}(\sigma H_2) = \sigma(23)H_1, \quad (2.177)$$

and then

$$\tau_1^{-1}(aeH_2 + b(12)H_2 + c(23)H_2) = a(23)H_1 + b(12)(23)H_1 + c(23)(23)H_1 \quad (2.178)$$

$$= a(23)H_1 + b(132)H_1 + ceH_1 \quad (2.179)$$

$$= a(23)H_1 + b(13)H_1 + ceH_1 \in \mathfrak{S}_3/H_1. \quad (2.180)$$

Thus, τ_1 is a linear isomorphism.

Definition of τ_2 . We define

$$\tau_2 : \mathbb{C}\{\mathfrak{S}_3/H_2\} \rightarrow \mathbb{C}\{\mathfrak{S}_3/H_3\} \quad (2.181)$$

by

$$\tau_2(\sigma H_2) = \sigma(12)H_3. \quad (2.182)$$

Then

$$\tau_2(aeH_2 + b(12)H_2 + c(23)H_2) = a\tau_2(eH_2) + b\tau_2((12)H_2) + c\tau_2((23)H_2) \quad (2.183)$$

$$= a(12)H_3 + b(12)(12)H_3 + c(23)(12)H_3 \quad (2.184)$$

$$= a(12)H_3 + b(12)(12)H_3 + c(123)H_3 \quad (2.185)$$

$$= a(12)H_3 + beH_3 + c(13)H_3. \quad (2.186)$$

Inverse of τ_2 . Define

$$\tau_2^{-1}(\sigma H_3) = \sigma(12)H_2, \quad (2.187)$$

and extend linearly. Then

$$\tau_2^{-1}(aeH_3 + b(12)H_3 + c(13)H_3) = a(12)H_2 + b(12)(12)H_2 + c(13)(12)H_2 \quad (2.188)$$

$$= a(12)H_2 + beH_2 + c(23)H_2. \quad (2.189)$$

Thus τ_2 is a linear isomorphism. Now that we have established that τ_1 and τ_2 are linear isomorphisms, we will check if $\rho_2(\sigma) \circ \tau_1 = \tau_1 \circ \rho_1(\sigma)$, and $\rho_3(\sigma) \circ \tau_2 = \tau_2 \circ \rho_2(\sigma)$ for all $\sigma \in \mathfrak{S}_3$. The first composition looks like this if we consider $aeH_1 + b(13)H_1 + c(23)H_1 \in \mathbb{C}\{\mathfrak{S}_3/H_1\}$:

$$\rho_2(\sigma)\tau_1(aeH_1 + b(13)H_1 + c(23)H_1) = \varphi_2(\sigma)(a(23)H_2 + b(12)H_2 + ceH_2) \quad (2.190)$$

$$= \sigma \cdot (a(23)H_2 + b(12)H_2 + ceH_2) \quad (2.191)$$

$$= a\sigma(23)H_2 + b\sigma(12)H_2 + c\sigma eH_2 \quad (2.192)$$

$$= a\sigma(23)H_2 + b\sigma(132)H_2 + c\sigma(23)(23)H_2 \quad (2.193)$$

$$= a\sigma(23)H_2 + b\sigma(13)(23)H_2 + c\sigma(23)(23)H_2 \quad (2.194)$$

$$= a\tau_1(\sigma eH_1) + b\tau_1(\sigma(13)H_1) + c\tau_1(\sigma(23)H_1) \quad (2.195)$$

$$= \tau_1(a\sigma eH_1) + \tau_1(b\sigma(13)H_1) + \tau_1(c\sigma(23)H_1) \quad (2.196)$$

$$= \tau_1(a\sigma eH_1 + b\sigma(13)H_1 + c\sigma(23)H_1) \quad (2.197)$$

$$= \tau_1(\sigma \cdot (aeH_1 + b(13)H_1 + c(23)H_1)) \quad (2.198)$$

$$= \tau_1\rho_1(\sigma)(aeH_1 + b(13)H_1 + c(23)H_1). \quad (2.199)$$

From this we can see that ρ_1 and ρ_2 are isomorphic. Moreover we will now consider some $aeH_2 + b(12)H_2 +$

$c(23)H_2 \in \mathbb{C}\{\mathfrak{S}_3/H_2\}$. Looking at the second composition we see that

$$\rho_3(\sigma)\tau_2(aeH_2 + b(12)H_2 + c(23)H_2) = \rho_3(g)(a(12)H_3 + beH_3 + c(13)H_3) \quad (2.200)$$

$$= \sigma \cdot (a(12)H_3 + beH_3 + c(13)H_3) \quad (2.201)$$

$$= a\sigma(12)H_3 + b\sigma eH_3 + c\sigma(13)H_3 \quad (2.202)$$

$$= a\sigma e(12)H_3 + b\sigma(12)(12)H_3 + c\sigma(123)H_3 \quad (2.203)$$

$$= a\sigma e(12)H_3 + b\sigma(12)(12)H_3 + c\sigma(23)(12)H_3 \quad (2.204)$$

$$= a\tau_2(\sigma eH_2) + b\tau_2(\sigma(12)H_2) + c\tau_2(\sigma(23)H_2) \quad (2.205)$$

$$= \tau_2(a\sigma eH_2) + \tau_2(b\sigma(12)H_2) + \tau_2(c\sigma(23)H_2) \quad (2.206)$$

$$= \tau_2(a\sigma eH_2 + b\sigma(12)H_2 + c\sigma(23)H_2) \quad (2.207)$$

$$= \tau_2(\sigma \cdot (aeH_2 + b(12)H_2 + c(23)H_2)) \quad (2.208)$$

$$= \tau_2\rho_c(\sigma)(aeH_2 + b(12)H_2 + c(23)H_2) \quad (2.209)$$

From this we can see that ρ_2 and ρ_3 are isomorphic. Since $\rho_1 \cong \rho_2$ and $\rho_2 \cong \rho_3$, we see that $\rho_1 \cong \rho_3$. $\square$

Problem 4. First we will let $H \leq \mathfrak{S}_n$. Consider the vector space $\mathbb{C}\{\underline{1}, \underline{2}, \dots, \underline{n}\} = \mathbb{C}[n]$. We can define the action of $\sigma \in \mathfrak{S}_n$ that takes $\underline{i} \in [n]$ and gives $\sigma \cdot \underline{i} = \sigma(\underline{i})$. We can also define the natural representation of $\mathfrak{S}_n$:

$$\varphi_1 : \mathfrak{S}_n \rightarrow \text{GL}(\mathbb{C}[n]) \quad (2.210)$$

With this construction we say that for $\sigma \in \mathfrak{S}_n$, and $\underline{i} \in [n]$, $\sigma \mapsto \varphi_1(\sigma)$ where $\varphi_1(\sigma)(\underline{i}) = \sigma \cdot \underline{i} = \underline{\sigma(i)}$. Furthermore,

$$\varphi_1(\sigma) \left(\sum_{i=1}^n \alpha_i \underline{i} \right) = \sigma \cdot \left(\sum_{i=1}^n \alpha_i \underline{i} \right) = \sum_{i=1}^n \alpha_i \underline{\sigma(i)}. \quad (2.211)$$

Now consider the coset representation on $\mathfrak{S}_n/H$:

$$\varphi_2 : \mathfrak{S}_n \rightarrow \text{GL}(\mathbb{C}\{\mathfrak{S}_n/H\}). \quad (2.212)$$

We say that for $\sigma \in \mathfrak{S}_n$, $\sigma \mapsto \varphi_2(\sigma)$. Furthermore, for $gH \in \mathfrak{S}_n/H$, we have $\varphi_2(\sigma)(gH) = \sigma \cdot (gH) = (\sigma g)H$. Now, we must find H such that there is a linear isomorphism, η , between the representations φ_1, φ_2 . This is show my showing, $\forall \sigma \in \mathfrak{S}_n$ that $\varphi_2(\sigma) \circ \eta = \eta \circ \varphi_1(\sigma)$. Consider $H = G_{\underline{i}}$ (the stabilizer) where $\underline{i} \in [n]$. We know that $|G_{\underline{i}}| = n - 1$ since $G_{\underline{i}}$ is exactly all permutations of $n - 1$ elements. Hence, $|\mathfrak{S}_n/G_{\underline{i}}| = n = |[n]|$. What this means is that $\mathbb{C}\{\mathfrak{S}_n/G_{\underline{i}}\}$ and $\mathbb{C}[n]$ have the same the number of basis elements. Now if we observe $\mathfrak{S}_n/G_{\underline{i}} = \{G_{\underline{i}}, \sigma_2 G_{\underline{i}}, \dots, \sigma_n G_{\underline{i}}\}$ we can define $\eta^{-1}(G_{\underline{i}}) = \underline{1}$, $\eta(\underline{1}) = eG_{\underline{i}}$, $\eta^{-1}(\sigma_j G_{\underline{i}}) = \underline{j}$, and $\eta(\underline{j}) = \sigma_j G_{\underline{i}}$ for $j = 2, 3, \dots, n$. We also do not choose our coset representatives arbitrarily. We require that $\sigma_j(\underline{i}) = \underline{j}$. This defines an isomorphism between $\mathfrak{S}_n/G_{\underline{i}}$ and $[n]$. Now in order to get a linear isomorphism we will consider arbitrary elements $\sum_{j=1}^n \beta_j \sigma_j G_{\underline{i}} \in \mathbb{C}\{\mathfrak{S}_n/G_{\underline{i}}\}$, and $\sum_{j=1}^n \beta_j \underline{j} \in \mathbb{C}[n]$. We define

$$\eta^{-1} \left(\sum_{j=1}^n \beta_j \sigma_j G_{\underline{i}} \right) = \sum_{j=1}^n \beta_j \eta^{-1}(\sigma_j G_{\underline{i}}) = \sum_{j=1}^n \beta_j \underline{j} \quad (2.213)$$

and

$$\eta \left(\sum_{j=1}^n \beta_j \underline{j} \right) = \sum_{j=1}^n \beta_j \eta(\underline{j}) = \sum_{j=1}^n \beta_j \sigma_j G_{\underline{i}}, \quad (2.214)$$

where $\sigma_1 = e$. Now we see that by definition η defines a linear isomorphism. Finally, we will check that $\varphi_2(\sigma) \circ \eta = \eta \circ \varphi_1(\sigma)$. We will reconsider our arbitrary element in $\mathbb{C}[n]$:

$$\varphi_2(\sigma) \eta \left(\sum_{j=1}^n \beta_j \underline{j} \right) = \varphi_2(\sigma) \left(\sum_{j=1}^n \beta_j \sigma_j G_{\underline{i}} \right) \quad (2.215)$$

$$= \sum_{j=1}^n \beta_j (\sigma \sigma_j) G_{\underline{i}} \quad (2.216)$$

We see that $\sigma\sigma_i$ sends $\underline{i}$ to $\underline{\sigma(j)}$ Hence $(\sigma\sigma_j)G_{\underline{i}} = \sigma_{\sigma_j}G_{\underline{i}}$

$$\sum_{j=1}^n \beta_j(\sigma\sigma_j)G_{\underline{i}} = \sum_{j=1}^n \beta_j\sigma_{\underline{\sigma(j)}}G_{\underline{i}} \quad (2.217)$$

Now we will test the other composition:

$$\eta\varphi_1(\sigma) \left(\sum_{j=1}^n \beta_j \underline{j} \right) = \eta \left(\sum_{j=1}^n \beta_j \underline{\sigma(j)} \right) \quad (2.218)$$

$$= \sum_{j=1}^n \beta_j \eta(\underline{\sigma(j)}) \quad (2.219)$$

$$= \sum_{j=1}^n \beta_j \sigma_{\underline{\sigma(j)}} G_{\underline{i}} \quad (2.220)$$

So, from this we can see we get a $\mathfrak{S}_n$ -isomorphism. $\square$

Problem 5. We will assume for the sake of contradiction that the regular representation of $\mathfrak{S}_3$ is isomorphic to the conjugation representation of $\mathfrak{S}_3$. We construct the left regular representation by the left multiplication action $\mathfrak{S}_3 \times \mathfrak{S}_3 \rightarrow \mathfrak{S}_3$ where $(\sigma, \tau) \mapsto \sigma \cdot \tau = \sigma\tau$ where $\sigma, \tau \in \mathfrak{S}_3$. The representation is defined by

$$\rho_1 : \mathfrak{S}_3 \rightarrow \text{GL}(\mathbb{C}\mathfrak{S}_3) \quad (2.221)$$

where $\rho_1(\tau) \left(\sum_{\sigma \in \mathfrak{S}_3} a_\sigma \sigma \right) = \sum_{\sigma \in \mathfrak{S}_3} a_\sigma (\tau\sigma)$. Furthermore, we know that $\dim \mathbb{C}\mathfrak{S}_3 = |\mathfrak{S}_3| = 3! = 6$. In a similar manner, we can construct the conjugation representation. We first take the action $\mathfrak{S}_3 \times \mathfrak{S}_3 \rightarrow \mathfrak{S}_3$ where $(\sigma, \tau) \mapsto \sigma \cdot \tau = \sigma\tau\sigma^{-1}$ and $\sigma, \tau \in \mathfrak{S}_3$. The representation is defined as follows

$$\rho_2 : \mathfrak{S}_3 \rightarrow \text{GL}(\mathbb{C}\mathfrak{S}_3). \quad (2.222)$$

So we see that the representation space is $\mathbb{C}\mathfrak{S}_3$. Since these two representations are isomorphic we know that it must be the case that $(\mathbb{C}\mathfrak{S}_3, \rho_1) \cong (\mathbb{C}\mathfrak{S}_3, \rho_2)$ as $\mathfrak{S}_3$ -modules. So it must be the case that $\chi_{\rho_1}(\sigma) = \chi_{\rho_2}(\sigma), \forall \sigma \in \mathfrak{S}_3$. We will check this claim. We know that

$$\chi_{\rho_2}(\sigma) = |Z_{\mathfrak{S}_3}(\sigma)| = \frac{|\mathfrak{S}_3|}{|K_\sigma|} \quad (2.223)$$

where K_g is the orbit of σ under the conjugation action of G on itself. Furthermore, we know that

$$\chi_{\rho_1}(\sigma) = \begin{cases} |G| & \text{if } g = e \\ 0 & \text{if } g \neq e \end{cases} \quad (2.224)$$

Now, consider the element $(12) \in \mathfrak{S}_3$. We will evaluate for the character of this element for each of the respective representations. We see

$$\chi_{\rho_2}((12)) = |Z_{\mathfrak{S}_3}((12))| = |\{e, (12)\}| = 2, \quad (2.225)$$

and we also see since $(12) \neq e$

$$\chi_{\rho_1}((12)) = 0. \quad (2.226)$$

This is a contradiction since $\chi_{\rho_1}((12)) = 0 \neq 2 = \chi_{\rho_2}((12))$, but for an isomorphism between ρ_1 and ρ_2 we require that these two values are equal. $\square$

Problem 6. Let M be an R -module. By definition we know that a subset $N \subseteq M$ is an R -submodule of M if N is an abelian subgroup of $(M, +)$ and $rn \in N, \forall r \in R, n \in N$. We will prove problem 5 in two parts

1. $(N_1 + N_2)$: Let N_1, N_2 be R -submodules of M . We know that $N_1 + N_2 := \{n_1 + n_2 \in M \mid n_1 \in N_1, n_2 \in N_2\}$. First we will show that $N_1 + N_2$ is an abelian subgroup of $(M, +)$. Take $n_1 + n_2, p_1 + p_2 \in N_1 + N_2$. We show closure

$$\textbf{Closure: } (n_1 + n_2) + (p_1 + p_2) = n_1 + n_2 + p_1 + p_2 \quad (2.227)$$

$$= n_1 + n_2 + p_2 + p_1 \quad (2.228)$$

$$= n_1 + (n_2 + p_2) + p_1 \quad (2.229)$$

$$= n_1 + p_1 + (n_2 + p_2) \quad (2.230)$$

$$= (n_1 + p_1) + (n_2 + p_2) \in N_1 + N_2 \quad (2.231)$$

We also show identity elements exist in the set. Since N_1, N_2 are both R -submodules, we know the identity exists in them respectively. So take $e + e \in N_1 + N_2$:

$$\textbf{Identity: } e + e = e \in N_1 + N_2. \quad (2.232)$$

Now we will show inverses. Since N_1, N_2 are R -submodules they contain inverses of each of their elements. So consider $n_1 + n_2 \in N_1 + N_2$. We will show that its inverse is contained within the sum. We will test if $n_1^{-1} + n_2^{-1} \in N_1 + N_2$ is the inverse

$$\textbf{Inverses: } (n_1 + n_2) + (n_1^{-1} + n_2^{-1}) = n_1 + n_2 + n_1^{-1} + n_2^{-1} \quad (2.233)$$

$$= n_1 + n_1^{-1} + n_2 + n_2^{-1} \quad (2.234)$$

$$= e + e \quad (2.235)$$

$$= e. \quad (2.236)$$

Here we see that inverses also exist within this set. Finally, we will show that this subgroup of $(M, +)$ is abelian. Once again, consider $n_1 + n_2, p_1 + p_2 \in N_1 + N_2$

$$\textbf{Commutativity: } (n_1 + n_2) + (p_1 + p_2) = n_1 + n_2 + p_1 + p_2 \quad (2.237)$$

$$= n_1 + p_1 + n_2 + p_2 \quad (2.238)$$

$$= n_1 + p_1 + p_2 + n_2 \quad (2.239)$$

$$= p_1 + n_1 + p_2 + n_2 \quad (2.240)$$

$$= p_1 + p_2 + n_1 + n_2 \quad (2.241)$$

$$= (p_1 + p_2) + (n_1 + n_2). \quad (2.242)$$

From these results we can say that $N_1 + N_2$ is an abelian subgroup of $(M, +)$. Now finally we must show that $r(n_1 + n_2) \in N_1 + N_2, \forall r \in R, n_1 + n_2 \in N_1 + N_2$. To preface, we know that $rn_1 \in N_1$ and $rn_2 \in N_2$. So when we evaluate $r(n_1 + n_2) = rn_1 + rn_2$ we arrive at this since these numbers live in our R -module M . This shows that $r(n_1 + n_2) = rn_1 + rn_2 \in N_1 + N_2$.

2. $(N_1 \cap N_2)$: Since N_1 and N_2 are R -submodules of M , we know that they are abelian subgroups of $(M, +)$. This means that $N_1 \cap N_2$ is also an abelian subgroup of $(M, +)$. With this in mind it suffices to show that for some $n \in N_1 \cap N_2, rn \in N_1 \cap N_2, \forall r \in R, n \in N_1 \cap N_2$. However, since $n \in N_1 \cap N_2$ means that $n \in N_1$ and $n \in N_2$, the result immediately follows.

□

Problem 7.

□

Problem 8. We will consider the cyclic group $\mathbb{Z}_n = \langle g \rangle$ of order n . Assume for the sake of contradiction that the n one-dimensional representations $\rho_q(g) = e^{2\pi i q/n}$ for $q = 0, 1, \dots, n-1$ are pairwise isomorphic. Moreover, we will say that each of these are equal functions. So if we consider a $q \neq r$ where $0 \leq q \leq n-1, 0 \leq r \leq n-1$, and $q > r$, we can see that

$$\rho_q(g) = e^{2\pi i q/n} = e^{2\pi i r/n} = \rho_r(g) \quad (2.243)$$

Moreover, we see that

$$e^{2\pi i q/n} = e^{2\pi i r/n} \quad (2.244)$$

$$e^{2\pi i(q-r)/n} = 1 \quad (2.245)$$

This result implies that $n|q-r$. Since $q, r \in [0, n-1]$ and are distinct integers, it is impossible for this to occur since $0 < q-r < n$. Here we arrive at a contradiction. This implies that this family of representations are not pairwise isomorphic. $\square$

Problem 9. We can remember that if we are given a set G , a finite set X , and a permutations representation of G , $V = \mathbb{C}X$, we can create the following map:

$$\rho : G \rightarrow \text{GL}_n(\mathbb{C}) \quad (2.246)$$

We say that G acts on the finite set X . Moreover, this causes the basis of $\mathbb{C}X$ to be $X = \{x_1, x_2, \dots, x_n\}$. If we pick an element $g \in G$. We can find the matrix representation of a particular permutation representation for g . This gives us a matrix $M(g)$ that is also a permutation matrix. Essentially, this matrix encodes a permutation of elements in our set X . This means that its entry at (i, j) is 1 if we have that $g \cdot x_j = x_i$, zeroes if this is not the case. Now if we are to consider the diagonal entries of $M(g)$, we see that $(M(g))_{ii} = 1$ if $g \cdot x_i = x_i$ and 0 in any other case. Since we know that

$$\chi^{\mathbb{C}X}(g) = \text{Tr}(M(g)), \quad (2.247)$$

we can actually say that

$$\chi^{\mathbb{C}X}(g) = \text{Tr}(M(g)) = |\{x \in X | g \cdot x = x\}| = |\text{Fix}(g)|, \forall g \in G \quad (2.248)$$

$\square$

Proof. Problem 10 Let V be a G -module with $\dim V = n$.

- (a) ($\chi^V(e) = n$): First we will consider a basis β for the G -module V . We know that $|\beta| = n = \dim V$. For a particular element $g \in G$ acting as a permutation, we require that the matrix representation of this is an $n \times n$ permutation matrix. Furthermore if we consider $M_\beta(e)$, we know from problem 9 that

$$\chi^V(e) = \text{Tr}(M_\beta(e)) = |\text{Fix}(e)|. \quad (2.249)$$

However, since we know that e fixes every element of G , it is the case that $|\text{Fix}(e)| = n$. So we get that

$$\chi^V(e) = \text{Tr}(M_\beta(e)) = |\text{Fix}(e)| = n \quad (2.250)$$

- (b) ($\chi^V(g) = \chi^V(h)$): We can remember that

$$\chi^V(h) = \text{Tr}(M_\beta(h)). \quad (2.251)$$

However, since $h = aga^{-1}$, and since we know that the trace is conjugation invariant, we get

$$\chi^V(h) = \text{Tr}(M_\beta(h)) = \text{Tr}(M_\beta(aga^{-1})) = \text{Tr}(M_\beta(g)) = \chi^V(g) \quad (2.252)$$

$\square$

Result 2.2 (Problem 11). Since we know that $\mathbb{Z}_4$ is abelian, we can say that each conjugacy class has one element. If we look at $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$. Furthermore this means that the columns of our table are each of the elements of our character table is

$$[0] \quad [1] \quad [2] \quad [3] \quad (2.253)$$

Since our group is abelian, we can consider the permutation φ and the fact that $[1]^4 = [0]$. From this we can conclude that $\varphi([1])^4 = 1$. There are three different potential values for $\varphi([1])$

$$\varphi([1]) = i \quad \varphi([1]) = -1 \quad \varphi([1]) = -i \quad (2.254)$$

$$\varphi([1])^2 = -1 \quad \varphi([1])^2 = 1 \quad \varphi([1])^2 = 1 \quad (2.255)$$

$$\varphi([1])^3 = -i \quad \varphi([1])^3 = -1 \quad \varphi([1])^4 = -1 \quad (2.256)$$

$$\varphi([1])^4 = 1 \quad \varphi([1])^4 = 1 \quad \varphi([1])^4 = 1. \quad (2.257)$$

Furthermore, since the representations are one dimensional, $\chi([1]^k) = \varphi([1]^k) = \varphi([1])^k$. This means that our character table looks something like this

	[1]	[2]	[3]	[4]
φ_1	1	1	1	1
φ_2	1	i	-1	$-i$
φ_3	1	-1	1	-1
φ_4	1	$-i$	-1	i

(2.258)

Now we will create the character table for $\mathbb{Z}_2 \times \mathbb{Z}_2$. First we will observe the group $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. We can first observe that

$$(0, 1)^2 = (0, 0) \quad (2.259)$$

$$(1, 0)^2 = (0, 0). \quad (2.260)$$

We can suppose we have one of our one-dimensional irreducible representations φ . We can see that

$$1 = \varphi((0, 0)) = \varphi((0, 1)^2) = \varphi((0, 1))^2 = 1^2 = 1 \quad (2.261)$$

$$1 = \varphi((0, 0)) = \varphi((1, 0)^2) = \varphi((1, 0))^2 = 1^2 = 1. \quad (2.262)$$

This tells us that $\varphi((0, 1)) = 1, -1$ and $\varphi((1, 0)) = 1, -1$. Moreover, since each of the irreducible representations are one-dimensional, we get for $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$ $\chi(g) = \varphi(g)$. Hence our character table will look something like this:

	(0, 0)	(0, 1)	(1, 0)	(1, 1)
φ_1	1	1	1	?
φ_2	1	-1	1	?
φ_3	1	1	-1	?
φ_4	1	-1	-1	?

(2.263)

However, since we are in fact working with 1-dimensional representations we get $\chi(gh) = \chi(g)\chi(h)$ for $g, h \in \mathbb{Z}_2 \times \mathbb{Z}_2$. This allows us to fill in the rest of the table:

	(0, 0)	(0, 1)	(1, 0)	(1, 1)
φ_1	1	1	1	1
φ_2	1	-1	1	-1
φ_3	1	1	-1	-1
φ_4	1	-1	-1	1

(2.264)

Problem 12. We will show that for the $\mathbb{C}$ -vector space $\mathcal{F}_G = \{f : G \rightarrow \mathbb{C}\}$ that the function

$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)} \quad (2.265)$$

is in fact an inner product of $\mathcal{F}_G$.

(a) **Conjugate Symmetry:** Consider the following for $\beta, \alpha \in \mathcal{F}_G$:

$$\overline{\langle \beta, \alpha \rangle} = \frac{1}{|G|} \sum_{g \in G} \beta(g) \overline{\alpha(g)} \quad (2.266)$$

$$= \frac{1}{|G|} \sum_{g \in G} \overline{\beta(g) \alpha(g)} \quad (2.267)$$

$$= \frac{1}{|G|} \sum_{g \in G} \overline{\beta(g)} \alpha(g) \quad (2.268)$$

$$= \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)} \quad (2.269)$$

$$= \langle \alpha, \beta \rangle \quad (2.270)$$

(b) **Linearity in the first argument:** Consider the following chain of equalities for $\alpha_1, \alpha_2, \beta \in \mathcal{F}_G$ and $c_1, c_2 \in \mathbb{C}$

$$c_1 \langle \alpha_1, \beta \rangle + c_2 \langle \alpha_2, \beta \rangle = c_1 \frac{1}{|G|} \sum_{g \in G} \alpha_1(g) \overline{\beta(g)} + c_2 \frac{1}{|G|} \sum_{g \in G} \alpha_2(g) \overline{\beta(g)} \quad (2.271)$$

$$= \frac{1}{|G|} \sum_{g \in G} c_1 \alpha_1(g) \overline{\beta(g)} + \frac{1}{|G|} \sum_{g \in G} c_2 \alpha_2(g) \overline{\beta(g)} \quad (2.272)$$

$$= \frac{1}{|G|} \sum_{g \in G} (c_1 \alpha_1(g) \overline{\beta(g)} + c_2 \alpha_2(g) \overline{\beta(g)}) \quad (2.273)$$

$$= \frac{1}{|G|} \sum_{g \in G} (c_1 \alpha_1(g) + c_2 \alpha_2(g)) \overline{\beta(g)} \quad (2.274)$$

$$= \langle c_1 \alpha_1 + c_2 \alpha_2, \beta \rangle \quad (2.275)$$

(c) **Positive definiteness:** First we will consider the function $\alpha \in \mathcal{F}_G$. So we get the following inequalities

$$\langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\alpha(g)} \quad (2.276)$$

We know that for any complex number z , $z\bar{z} = |z|^2$. This means that $z\bar{z} = |z|^2 \geq 0$ regardless of the choice of z . Hence

$$\langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\alpha(g)} \quad (2.277)$$

$$= \frac{1}{|G|} \sum_{g \in G} |\alpha(g)|^2 \quad (2.278)$$

$$\geq 0 \quad (2.279)$$

Since the sum of non-negative terms is non-negative we get this result. Moreover, if $\alpha = 0$, this means that $\alpha(g) = 0$ for all $g \in G$. So if $\alpha = 0$

$$\langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} |\alpha(g)|^2 \quad (2.280)$$

$$= \frac{1}{|G|} \sum_{g \in G} |0|^2 \quad (2.281)$$

$$= 0 \quad (2.282)$$

Now if we first let $\langle \alpha, \alpha \rangle = 0$, we see that

$$0 = \langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} |\alpha(g)|^2. \quad (2.283)$$

However, we see that this is only true when $\alpha(g) = 0$ for all $g \in G$. This condition implies that $\alpha = 0$

□

Result 2.3 (Problem 13). We say that a G -module V where $V \neq \{0\}$ if its only G -submodules are $\{0\}$ and itself.

Result 2.4 (Problem 14). In order to answer this question we must first consider the conjugacy classes of $\mathfrak{S}_3$. We can see that there are only three of them

$$\{e\} \quad \{(12), (13), (23)\} \quad \{(123), (132)\} \quad (2.284)$$

Since the number of irreducible representations of $\mathfrak{S}_3$ is the same as the number of conjugacy classes, we can say that these irreducible representations are the only ones of the group up to isomorphism.

Proof. Let $R(G)$ denote the set of all class functions on G . We will show that $R(G)$ is a subspace of $\mathcal{F}_G$.

- (a) **Zero Function:** We will consider the zero function $0 : G \rightarrow \mathbb{C}$. We know that this function maps any $k \in G$ to 0. Hence we see that

$$0(hgh^{-1}) = 0 = 0(g). \quad (2.285)$$

This shows us that 0 is a class function.

- (b) **Closure under addition:** Consider the two class functions φ_1 , and φ_2 . Since they are class functions we know that

$$\varphi_1(hgh^{-1}) = \varphi_1(g) \quad (2.286)$$

$$\varphi_2(hgh^{-1}) = \varphi_2(g) \quad (2.287)$$

$\forall g, h \in G$. Moreover, we can see that

$$(\varphi_1 + \varphi_2)(hgh^{-1}) = \varphi_1(hgh^{-1}) + \varphi_2(hgh^{-1}). \quad (2.288)$$

This allows us to see that

$$(\varphi_1 + \varphi_2)(hgh^{-1}) = \varphi_1(hgh^{-1}) + \varphi_2(hgh^{-1}) \quad (2.289)$$

$$= \varphi_1(g) + \varphi_2(g) \quad (2.290)$$

$$= (\varphi_1 + \varphi_2)(g). \quad (2.291)$$

Hence, we see that $(\varphi_1 + \varphi_2) \in R(G)$

- (c) **Scalar Multiplication:** If we consider a $\varphi \in R(G)$ and a $z \in \mathbb{C}$, we can see that

$$(c\varphi)(hgh^{-1}) = c\varphi(hgh^{-1}) = c\varphi(g) = (c\varphi)(g). \quad (2.292)$$

This shows us that scalar multiplications preserve class functions.

□

2.4 Gonzalez's Answers for Homework 3

1. Compute the matrices $M(\sigma)$ for all $\sigma \in \mathfrak{S}_3$ in the natural representation. Compute the traces of these matrices, and see if there is some pattern in the traces.

Solution:

Recall that in the natural representation of $\mathfrak{S}_3$ on $\mathbb{C}[3] = \mathbb{C}\{\underline{1}, \underline{2}, \underline{3}\}$, the action is $\sigma \cdot \underline{i} = \underline{\sigma(i)}$. In the basis $(\underline{1}, \underline{2}, \underline{3})$, the matrix $M(\sigma)$ is the permutation matrix whose j -th column is $e_{\sigma(j)}$, i.e. the entry $(M(\sigma))_{i,j} = \delta_{\sigma(j),i}$.

Listing all six elements:

$$M(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M((12)) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M((13)) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$M((23)) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad M((123)) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad M((132)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

The traces are

$$\text{Tr}M(e) = 3, \quad \text{Tr}M((12)) = \text{Tr}M((13)) = \text{Tr}M((23)) = 1, \quad \text{Tr}M((123)) = \text{Tr}M((132)) = 0.$$

Pattern. For each σ , $\text{Tr}M(\sigma) = |\text{Fix}(\sigma)| = \#\{i \in [3] : \sigma(i) = i\}$. Indeed, the diagonal entry $(M(\sigma))_{i,i} = \delta_{\sigma(i),i}$ equals 1 exactly when i is fixed by σ . Moreover, the trace is constant on conjugacy classes: it depends only on the cycle type of σ (the identity has 3 fixed points, transpositions have 1, and 3-cycles have 0).

2. Show that the sign representation $\text{sgn}: \mathfrak{S}_n \rightarrow \mathbb{C}^*$ is not a permutation representation.

Solution:

Suppose, towards a contradiction, that the sign representation is a permutation representation. By definition, this means there is a finite set X on which $\mathfrak{S}_n$ acts and a $\mathfrak{S}_n$ -equivariant linear isomorphism $\varphi: \mathbb{C}X \rightarrow \mathbb{C}$. In particular, $\dim \mathbb{C}X = \dim \mathbb{C} = 1$, so $|X| = 1$, say $X = \{x_0\}$. Since the only function from $\mathfrak{S}_n$ to permutations of a one-point set is trivial, we must have $\sigma \cdot x_0 = x_0$ for all $\sigma \in \mathfrak{S}_n$. Hence the action of $\mathfrak{S}_n$ on $\mathbb{C}X$ is the trivial representation, and so sgn would have to equal the trivial character, i.e. $\text{sgn}(\sigma) = 1$ for all σ .

But for $n \geq 2$, $\text{sgn}((12)) = -1 \neq 1$, a contradiction. Therefore sgn is not a permutation representation.

(Equivalent formulation: in a permutation representation the matrix of every group element is a permutation matrix, in particular has nonnegative real entries. For

$n \geq 2$, the 1×1 “matrix” of (12) in the sign representation is $[-1]$, which is not a permutation matrix.)

3. Show that if $H_1 = \langle (12) \rangle$, $H_2 = \langle (13) \rangle$, $H_3 = \langle (23) \rangle$, then the representations on $\mathfrak{S}_3/H_1$, $\mathfrak{S}_3/H_2$, and $\mathfrak{S}_3/H_3$ are all isomorphic.

Solution:

We use the fact: if $H, K \leq G$ $K = gHg^{-1}$ for some $g \in G$, if and only if the coset G -sets G/H and G/K are isomorphic.

After linearization, this G -set isomorphism induces an isomorphism of permutation representations $\mathbb{C}\{G/H\} \cong_G \mathbb{C}\{G/K\}$.

It remains to show that H_1, H_2, H_3 are pairwise conjugate in $\mathfrak{S}_3$. This follows because all transpositions in $\mathfrak{S}_n$ form a single conjugacy class. Concretely:

$$(13)(12)(13)^{-1} = (32) = (23), \quad (23)(12)(23)^{-1} = (13).$$

Hence $(13)H_1(13)^{-1} = H_3$ and $(23)H_1(23)^{-1} = H_2$. By the general fact above,

$$\mathfrak{S}_3/H_1 \cong_{\mathfrak{S}_3} \mathfrak{S}_3/H_2 \cong_{\mathfrak{S}_3} \mathfrak{S}_3/H_3,$$

and the same isomorphisms persist after passing to the corresponding permutation representations.

4. Find a subgroup $H \leq \mathfrak{S}_n$ such that the representation on $\mathfrak{S}_n/H$ is isomorphic to the natural representation of $\mathfrak{S}_n$.

Solution:

Take $H = \text{St}_{\mathfrak{S}_n}(\underline{n}) = \{\sigma \in \mathfrak{S}_n : \sigma(n) = n\} \cong \mathfrak{S}_{n-1}$, the stabilizer of the point n in the natural action of $\mathfrak{S}_n$ on $[n] = \{1, 2, \dots, n\}$.

The natural action of $\mathfrak{S}_n$ on $[n]$ is transitive, so by the orbit–stabilizer theorem, the map

$$\Psi: \mathfrak{S}_n/H \longrightarrow [n], \quad \sigma H \mapsto \sigma(n)$$

is a well-defined $\mathfrak{S}_n$ -equivariant bijection. Indeed, if $\sigma H = \sigma' H$ then $\sigma' = \sigma h$ for some $h \in H$, and $\sigma'(n) = \sigma(h(n)) = \sigma(n)$; equivariance is immediate from $(\tau\sigma)(n) = \tau(\sigma(n))$; surjectivity is the transitivity of the action; and the two finite sets have the same cardinality n .

Linearizing Ψ gives a $\mathfrak{S}_n$ -isomorphism between the coset representation $\mathbb{C}\{\mathfrak{S}_n/H\}$ and the natural representation $\mathbb{C}[n]$.

5. Show that the regular representation of $\mathfrak{S}_3$ is *not* isomorphic to the conjugation representation of $\mathfrak{S}_3$ (where G acts on $\mathbb{C}G$ by $g \cdot h = ghg^{-1}$).

Solution:

First solution using a G -isomorphism. Suppose there exists a linear isomorphism $\phi : \mathbb{C}\mathfrak{S}_3^{\text{conj}} \rightarrow \mathbb{C}\mathfrak{S}_3^{\text{reg}}$ such that for all $\sigma, \tau \in \mathfrak{S}_3$,

$$\phi(\sigma \cdot \tau) = \sigma \cdot \phi(\tau).$$

According to the actions on the two representations, this means that for all $\sigma, \tau \in \mathfrak{S}_3$ we must have

$$\phi(\sigma\tau\sigma^{-1}) = \sigma\phi(\tau).$$

If we choose an element $x \in \mathbb{C}\mathfrak{S}_3^{\text{conj}}$ such that for all $\sigma \in \mathfrak{S}_3$,

$$x = \sigma x \sigma^{-1}, \tag{1}$$

then we must have

$$\phi(x) = \phi(\sigma x \sigma^{-1}) = \sigma \phi(x).$$

This implies that if

$$\phi(x) = a_e e + a_{(12)}(12) + a_{(13)}(13) + a_{(23)}(23) + a_{(123)}(123) + a_{(132)}(132),$$

then $a_e = a_{(12)} = a_{(13)} = a_{(23)} = a_{(123)} = a_{(132)}$, since the regular action on $\mathfrak{S}_3$ has a unique orbit (it is transitive). We would then have

$$\phi(x) = c_x (e + (12) + (13) + (23) + (123) + (132)) = c_x \sum_{\tau \in \mathfrak{S}_3} \tau.$$

Now note that the elements $e, (123) + (132), (12) + (13) + (23) \in \mathbb{C}\mathfrak{S}_3^{\text{conj}}$ all satisfy equation (1), since they are sums over the elements of the 3 different orbits of the conjugation action on $\mathfrak{S}_3$.

By the discussion above, if $\phi(e) = c_e \sum_{\tau \in \mathfrak{S}_3} \tau$ and $\phi((123) + (132)) = c_{(123)+(132)} \sum_{\tau \in \mathfrak{S}_3} \tau$, then

$$\phi(c_{(123)+(132)} e - c_e((123) + (132))) = 0.$$

This means there is a nonzero element $c_{(123)+(132)} e - c_e((123) + (132)) \neq 0$ in $\ker \phi$, so ϕ cannot be an isomorphism, a contradiction.

Another solution using characters. We compute the characters of both representations and observe that they differ. Since isomorphic representations have equal characters, this suffices.

Both representations have $G = \mathfrak{S}_3$ acting on the vector space $\mathbb{C}G$, which has basis G itself. In each case, the basis is permuted by the action, so each is a permutation

representation, and Problem 9 below tells us that the character at g counts fixed basis elements.

Regular representation. For $g \in G$, the fixed elements of the basis are $\{h \in G : gh = h\} = \{h : g = e\}$. Hence

$$\chi^{\text{reg}}(g) = \begin{cases} |G| = 6 & \text{if } g = e, \\ 0 & \text{otherwise.} \end{cases}$$

On the three conjugacy classes $\{e\}, \{(12), (13), (23)\}, \{(123), (132)\}$, we get $\chi^{\text{reg}} = (6, 0, 0)$.

Conjugation representation. For $g \in G$, the fixed basis elements are $\{h \in G : ghg^{-1} = h\} = C_G(g)$, the centralizer of g . So $\chi^{\text{conj}}(g) = |C_G(g)|$. For $\mathfrak{S}_3$ we have:

- $C_{\mathfrak{S}_3}(e) = \mathfrak{S}_3$, so $\chi^{\text{conj}}(e) = 6$;
- $C_{\mathfrak{S}_3}((12)) = \langle (12) \rangle$, so $\chi^{\text{conj}}((12)) = 2$ (similarly for $(13), (23)$);
- $C_{\mathfrak{S}_3}((123)) = \langle (123) \rangle$, so $\chi^{\text{conj}}((123)) = 3$ (similarly for (132)).

Thus on the three conjugacy classes, $\chi^{\text{conj}} = (6, 2, 3)$.

Since $\chi^{\text{reg}} \neq \chi^{\text{conj}}$ (e.g., $\chi^{\text{reg}}((12)) = 0 \neq 2 = \chi^{\text{conj}}((12))$), the two representations are not isomorphic.

6. Let M be an R -module and let N_1, N_2 be R -submodules. Show that $N_1 + N_2$ is an R -submodule of M , and that $N_1 \cap N_2$ is an R -submodule of M .

Solution:

Recall $N_1 + N_2 = \{n_1 + n_2 : n_1 \in N_1, n_2 \in N_2\} \subseteq M$.

$N_1 + N_2$ is a submodule.

- *Contains 0:* $0 = 0 + 0$ with $0 \in N_1$ and $0 \in N_2$.
- *Closed under addition:* For $n_1 + n_2, n'_1 + n'_2 \in N_1 + N_2$,

$$(n_1 + n_2) + (n'_1 + n'_2) = (n_1 + n'_1) + (n_2 + n'_2) \in N_1 + N_2,$$
since $n_1 + n'_1 \in N_1$ and $n_2 + n'_2 \in N_2$ by closure of each N_i .
- *Closed under scalar multiplication:* For $r \in R$ and $n_1 + n_2 \in N_1 + N_2$,

$$r(n_1 + n_2) = rn_1 + rn_2 \in N_1 + N_2,$$
since $rn_1 \in N_1$ and $rn_2 \in N_2$.

$N_1 \cap N_2$ is a submodule.

- *Contains 0:* $0 \in N_1$ and $0 \in N_2$, so $0 \in N_1 \cap N_2$.
- *Closed under addition:* If $x, y \in N_1 \cap N_2$, then $x, y \in N_1$ implies $x + y \in N_1$, and similarly $x + y \in N_2$, so $x + y \in N_1 \cap N_2$.
- *Closed under scalar multiplication:* If $r \in R$ and $x \in N_1 \cap N_2$, then $rx \in N_1$ and $rx \in N_2$, hence $rx \in N_1 \cap N_2$.

This completes the proof.

7. Show that the standard representation $V_2 = \mathbb{C}\{\underline{2} - \underline{1}, \underline{3} - \underline{1}\}$ of $\mathfrak{S}_3$ is irreducible.

Hint: If $W \subseteq V_2$ is a nonzero $\mathfrak{S}_3$ -submodule, show $\dim W = 2$ by considering the action of generators on a nonzero vector in W .

Solution:

Let $e_1 := \underline{2} - \underline{1}$ and $e_2 := \underline{3} - \underline{1}$, so $V_2 = \mathbb{C}\{e_1, e_2\}$ has dimension 2. Suppose $W \subseteq V_2$ is a nonzero $\mathfrak{S}_3$ -submodule. Since $\dim W \in \{1, 2\}$ and $W \subseteq V_2$, it suffices to rule out $\dim W = 1$.

Suppose, for contradiction, that $W = \mathbb{C}w$ for some nonzero $w = \alpha e_1 + \beta e_2$. Then for every $\sigma \in \mathfrak{S}_3$, the vector $\sigma \cdot w$ must be a scalar multiple of w . We will get a contradiction using the generators (12) and (23).

Action of (12). Since (12) swaps $\underline{1} \leftrightarrow \underline{2}$ and fixes $\underline{3}$,

$$\begin{aligned} (12) \cdot e_1 &= (12)(\underline{2} - \underline{1}) = \underline{1} - \underline{2} = -e_1, \\ (12) \cdot e_2 &= (12)(\underline{3} - \underline{1}) = \underline{3} - \underline{2} = (\underline{3} - \underline{1}) - (\underline{2} - \underline{1}) = -e_1 + e_2. \end{aligned}$$

Hence $(12) \cdot w = -\alpha e_1 + \beta(-e_1 + e_2) = -(\alpha + \beta)e_1 + \beta e_2$. The condition $(12) \cdot w \in \mathbb{C}w$ requires this to be a scalar multiple $\lambda w = \lambda \alpha e_1 + \lambda \beta e_2$:

$$-(\alpha + \beta) = \lambda \alpha \quad \text{and} \quad \beta = \lambda \beta. \quad (*)$$

Action of (23). Since (23) swaps $\underline{2} \leftrightarrow \underline{3}$ and fixes $\underline{1}$,

$$\begin{aligned} (23) \cdot e_1 &= (23)(\underline{2} - \underline{1}) = \underline{3} - \underline{1} = e_2, \\ (23) \cdot e_2 &= (23)(\underline{3} - \underline{1}) = \underline{2} - \underline{1} = e_1. \end{aligned}$$

So $(23) \cdot w = \alpha e_2 + \beta e_1 = \beta e_1 + \alpha e_2$. The condition $(23) \cdot w \in \mathbb{C}w$ requires $(\beta, \alpha) = \mu(\alpha, \beta)$ for some $\mu \in \mathbb{C}$:

$$\beta = \mu \alpha \quad \text{and} \quad \alpha = \mu \beta. \quad (**)$$

From (**): $\alpha = \mu\beta = \mu(\mu\alpha) = \mu^2\alpha$, so either $\alpha = 0$ or $\mu^2 = 1$. Symmetrically, either $\beta = 0$ or $\mu^2 = 1$.

Case 1: $\mu^2 = 1$, i.e. $\mu = \pm 1$. Then $\beta = \mu\alpha = \pm\alpha$, so w is a scalar multiple of $e_1 + e_2 = \underline{2} + \underline{3} - \underline{2}\underline{1}$ or of $e_1 - e_2 = \underline{2} - \underline{3}$. Plug into (*):

- If $\beta = \alpha$ (so $\mu = 1$), then $-(2\alpha) = \lambda\alpha$ and $\alpha = \lambda\alpha$, forcing $\alpha = 0$ (else $\lambda = -2$ and $\lambda = 1$ contradict).
- If $\beta = -\alpha$ (so $\mu = -1$), then $0 = \lambda\alpha$ and $-\alpha = \lambda(-\alpha)$, forcing $\alpha = 0$ (else $\lambda = 0$ and $\lambda = 1$ contradict).

In both subcases $\alpha = \beta = 0$, contradicting $w \neq 0$.

Case 2: $\alpha = 0$. Then $w = \beta e_2$ with $\beta \neq 0$. From (**): $\beta = \mu \cdot 0 = 0$, contradicting $\beta \neq 0$.

Case 3: $\beta = 0$. Symmetric: $w = \alpha e_1$ with $\alpha \neq 0$, but (**) gives $\alpha = \mu \cdot 0 = 0$.

All cases lead to a contradiction, so no 1-dimensional $\mathfrak{S}_3$ -submodule of V_2 exists. Hence $W = V_2$, and V_2 is irreducible.

8. Let $\mathbb{Z}_n = \langle g \rangle$ be the cyclic group of order n . Show that the n one-dimensional representations $\rho_q(g) = e^{2\pi i q/n}$ for $q = 0, 1, \dots, n-1$ are pairwise non-isomorphic.

Hint: Two one-dimensional representations are isomorphic if and only if they are equal as functions.

Solution:

By the hint, it is enough to verify that the functions $\rho_q: \mathbb{Z}_n \rightarrow \mathbb{C}^*$ are pairwise distinct for $q = 0, 1, \dots, n-1$. Since $\mathbb{Z}_n = \langle g \rangle$, a homomorphism is determined by its value on g , so it suffices to check $\rho_q(g) \neq \rho_p(g)$ when $0 \leq p < q \leq n-1$.

Suppose $\rho_q(g) = \rho_p(g)$, i.e. $e^{2\pi i q/n} = e^{2\pi i p/n}$. Then

$$e^{2\pi i(q-p)/n} = 1,$$

which forces $\frac{q-p}{n} \in \mathbb{Z}$, i.e. $n \mid (q-p)$. But $0 \leq q-p \leq n-1$, so the only possibility is $q-p = 0$, i.e. $q = p$. Therefore the maps $\rho_0, \rho_1, \dots, \rho_{n-1}$ are pairwise distinct as functions, hence pairwise non-isomorphic as representations.

9. Prove that if $V = \mathbb{C}X$ is a permutation representation of G on a finite set X , then $\chi^V(g) = |\{x \in X \mid g \cdot x = x\}|$ for all $g \in G$.

Hint: How are the matrices $M(g)$ defined in the permutation representation? What is the trace of such a matrix?

Solution:

Order $X = \{x_1, x_2, \dots, x_n\}$ and use this as a basis for $V = \mathbb{C}X$. By definition of the permutation representation, $g \cdot x_j = x_{\sigma(j)}$ where $\sigma \in \mathfrak{S}_X$ is the permutation of X induced by g . Therefore the matrix $M(g)$ in this basis has entries

$$(M(g))_{i,j} = \delta_{i,\sigma(j)} = \begin{cases} 1 & \text{if } g \cdot x_j = x_i, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, the diagonal entry $(M(g))_{i,i} = \delta_{i,\sigma(i)}$ equals 1 exactly when $g \cdot x_i = x_i$, and 0 otherwise. Summing the diagonal:

$$\chi^V(g) = \text{Tr} M(g) = \sum_{i=1}^n (M(g))_{i,i} = \sum_{i=1}^n \mathbf{1}[g \cdot x_i = x_i] = |\{x \in X : g \cdot x = x\}|.$$

10. Let V be a G -module with $\dim V = n$. Show:

(a) $\chi^V(e) = n$.

(b) If g and h are conjugate in G (i.e. $h = aga^{-1}$ for some $a \in G$), then $\chi^V(g) = \chi^V(h)$.

Hint for (b): trace is conjugation-invariant.

Solution:

(a) Since V is a G -module, the action of the identity $e \in G$ is the identity transformation $\text{id}_V \in \text{GL}(V)$. In any basis of V , the matrix of id_V is the $n \times n$ identity matrix I_n , and so

$$\chi^V(e) = \text{Tr} I_n = n.$$

(b) Fix a basis of V , and write $M(g)$ for the matrix of the action of g . Since V is a representation, $M: G \rightarrow \text{GL}_n(\mathbb{C})$ is a homomorphism, so

$$M(h) = M(aga^{-1}) = M(a) M(g) M(a)^{-1}.$$

Trace is invariant under conjugation in $\text{GL}_n(\mathbb{C})$ (because $\text{Tr}(AB) = \text{Tr}(BA)$ for all square matrices, hence $\text{Tr}(PXP^{-1}) = \text{Tr}(P^{-1}PX) = \text{Tr} X$). Therefore

$$\chi^V(h) = \text{Tr} M(h) = \text{Tr}(M(a)M(g)M(a)^{-1}) = \text{Tr} M(g) = \chi^V(g).$$

11. Construct the character tables of the following groups:

- (a) $\mathbb{Z}_4$ (four irreducible characters, use $i = e^{2\pi i/4}$).
 (b) $\mathbb{Z}_2 \times \mathbb{Z}_2$ (four conjugacy classes, four one-dimensional irreducibles).

Solution:

Both groups are abelian of order 4, so each element forms its own conjugacy class, giving 4 classes and hence 4 irreducible representations. For abelian groups all irreducible representations are 1-dimensional, so the irreducible characters are exactly the group homomorphisms $G \rightarrow \mathbb{C}^*$.

(a) $\mathbb{Z}_4 = \langle g \mid g^4 = e \rangle$. By Problem 8, the four irreducible characters are $\chi_q(g^k) = e^{2\pi i qk/4} = i^{qk}$ for $q = 0, 1, 2, 3$.

$\mathbb{Z}_4$	e	g	g^2	g^3
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

(b) $\mathbb{Z}_2 \times \mathbb{Z}_2 = \langle a, b \mid a^2 = b^2 = e, ab = ba \rangle$. Any homomorphism $\chi: \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \mathbb{C}^*$ is determined by $\chi(a), \chi(b) \in \{\pm 1\}$. The four characters are $\chi_{\varepsilon, \delta}(a^i b^j) = \varepsilon^i \delta^j$ for $\varepsilon, \delta \in \{\pm 1\}$.

$\mathbb{Z}_2 \times \mathbb{Z}_2$	e	a	b	ab
χ_{++}	1	1	1	1
χ_{-+}	1	-1	1	-1
χ_{+-}	1	1	-1	-1
χ_{--}	1	-1	-1	1

A quick orthogonality check confirms correctness: each row has norm-squared $\frac{1}{4}(1 + 1 + 1 + 1) = 1$, and any two distinct rows have inner product 0.

12. Let G be a finite group and let $\mathcal{F}_G = \{f: G \rightarrow \mathbb{C}\}$ be the $\mathbb{C}$ -vector space of all functions from G to $\mathbb{C}$. Define

$$\langle \alpha, \beta \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)}.$$

Show that $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{F}_G$. That is, verify:

- (a) **Conjugate symmetry:** $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$ for all $\alpha, \beta \in \mathcal{F}_G$.
 (b) **Linearity in the first argument:** $\langle c_1 \alpha_1 + c_2 \alpha_2, \beta \rangle = c_1 \langle \alpha_1, \beta \rangle + c_2 \langle \alpha_2, \beta \rangle$ for all $c_1, c_2 \in \mathbb{C}$ and $\alpha_1, \alpha_2, \beta \in \mathcal{F}_G$.

(c) **Positive definiteness:** $\langle \alpha, \alpha \rangle \geq 0$ for all α , with equality if and only if $\alpha = 0$.

Solution:

We use repeatedly the basic facts $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{z\bar{w}} = \bar{z}w$, $\overline{\bar{z}} = z$, and $z\bar{z} = |z|^2 \geq 0$ for $z \in \mathbb{C}$.

(a) **Conjugate symmetry.**

$$\overline{\langle \beta, \alpha \rangle} = \overline{\frac{1}{|G|} \sum_{g \in G} \beta(g) \overline{\alpha(g)}} = \frac{1}{|G|} \sum_{g \in G} \overline{\beta(g)} \cdot \overline{\overline{\alpha(g)}} = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\beta(g)} = \langle \alpha, \beta \rangle.$$

(We used that $1/|G|$ is a positive real, hence equals its own conjugate.)

(b) **Linearity in the first argument.** For $c_1, c_2 \in \mathbb{C}$ and $\alpha_1, \alpha_2, \beta \in \mathcal{F}_G$,

$$\begin{aligned} \langle c_1 \alpha_1 + c_2 \alpha_2, \beta \rangle &= \frac{1}{|G|} \sum_{g \in G} (c_1 \alpha_1(g) + c_2 \alpha_2(g)) \overline{\beta(g)} \\ &= c_1 \cdot \frac{1}{|G|} \sum_{g \in G} \alpha_1(g) \overline{\beta(g)} + c_2 \cdot \frac{1}{|G|} \sum_{g \in G} \alpha_2(g) \overline{\beta(g)} \\ &= c_1 \langle \alpha_1, \beta \rangle + c_2 \langle \alpha_2, \beta \rangle. \end{aligned}$$

(c) **Positive definiteness.**

$$\langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g) \overline{\alpha(g)} = \frac{1}{|G|} \sum_{g \in G} |\alpha(g)|^2.$$

Each summand $|\alpha(g)|^2 \geq 0$, and the prefactor $1/|G| > 0$, so $\langle \alpha, \alpha \rangle \geq 0$. Moreover, since this is a sum of nonnegative terms,

$$\langle \alpha, \alpha \rangle = 0 \iff |\alpha(g)|^2 = 0 \text{ for every } g \in G \iff \alpha(g) = 0 \text{ for every } g \iff \alpha = 0.$$

13. Let $V = \mathbb{C}\{\underline{2} - \underline{1}, \underline{3} - \underline{1}\}$ be the standard representation of $\mathfrak{S}_3$. Its character χ^V takes values:

$$\chi^V(e) = 2, \quad \chi^V((12)) = \chi^V((13)) = \chi^V((23)) = 0, \quad \chi^V((123)) = \chi^V((132)) = -1.$$

Verify that V is irreducible.

Hint: Compute $\langle \chi^V, \chi^V \rangle$.

Solution:

Recall the irreducibility criterion: a representation V of a finite group G is irreducible if and only if $\langle \chi^V, \chi^V \rangle = 1$. We compute this inner product directly. Since χ^V is constant on each conjugacy class, we can group the sum by classes, with sizes $|\{e\}| = 1$, $|\{(12), (13), (23)\}| = 3$, $|\{(123), (132)\}| = 2$:

$$\langle \chi^V, \chi^V \rangle = \frac{1}{|\mathfrak{S}_3|} \sum_{\sigma \in \mathfrak{S}_3} |\chi^V(\sigma)|^2 = \frac{1}{6} (1 \cdot |2|^2 + 3 \cdot |0|^2 + 2 \cdot |-1|^2) = \frac{1}{6} (4 + 0 + 2) = 1.$$

Therefore V is irreducible.

14. Show that the trivial representation $\mathbf{1}_{\mathfrak{S}_3}$, the sign representation $\text{sgn}_{\mathfrak{S}_3}$, and the standard representation V_3 are the *only* irreducible representations of $\mathfrak{S}_3$ (up to isomorphism).

Hint: How many conjugacy classes are there in $\mathfrak{S}_3$?

Solution:

Recall the fundamental theorem: the number of irreducible representations of a finite group G (up to isomorphism) equals the number of conjugacy classes of G .

Conjugacy classes of $\mathfrak{S}_3$. Two elements of $\mathfrak{S}_n$ are conjugate iff they have the same cycle type. The cycle types in $\mathfrak{S}_3$ are: $\{1, 1, 1\}$ (the identity), $\{2, 1\}$ (transpositions), $\{3\}$ (3-cycles). So $\mathfrak{S}_3$ has exactly 3 conjugacy classes:

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

Hence there are exactly 3 irreducible representations of $\mathfrak{S}_3$ up to isomorphism.

Three irreducibles already found. We exhibit three pairwise non-isomorphic irreducibles:

- $\mathbf{1}_{\mathfrak{S}_3}$: trivial, $\dim = 1$, character $(1, 1, 1)$.
- $\text{sgn}_{\mathfrak{S}_3}$: sign, $\dim = 1$, character $(1, -1, 1)$.
- V_3 : standard, $\dim = 2$, character $(2, 0, -1)$ (irreducible by Problem 13).

The first two are irreducible because they are 1-dimensional. They are pairwise non-isomorphic because their characters differ: $\mathbf{1}$ and sgn disagree at (12) , and V_3 has different dimension ($\chi(e) = 2$) from the other two ($\chi(e) = 1$).

Conclusion. Three pairwise non-isomorphic irreducibles in a list of length 3 exhausts all irreducible representations of $\mathfrak{S}_3$.

15. A function $f: G \rightarrow \mathbb{C}$ is a *class function* if $f(hgh^{-1}) = f(g)$ for all $g, h \in G$. Let $R(G)$ denote the set of all class functions on G . Show that $R(G)$ is a subspace of $\mathcal{F}_G$. That is, verify that $R(G)$ is closed under addition and scalar multiplication, and contains the zero function.

Solution:

We verify the three subspace axioms.

Zero function. The zero function $\mathbf{0}: G \rightarrow \mathbb{C}$, $\mathbf{0}(g) = 0$ for all g , satisfies $\mathbf{0}(hgh^{-1}) = 0 = \mathbf{0}(g)$ for all $g, h \in G$. Hence $\mathbf{0} \in R(G)$.

Closure under addition. Let $f_1, f_2 \in R(G)$ and $g, h \in G$. Then

$$(f_1 + f_2)(hgh^{-1}) = f_1(hgh^{-1}) + f_2(hgh^{-1}) = f_1(g) + f_2(g) = (f_1 + f_2)(g),$$

where the middle equality uses that each f_i is a class function. Hence $f_1 + f_2 \in R(G)$.

Closure under scalar multiplication. Let $f \in R(G)$, $c \in \mathbb{C}$, and $g, h \in G$. Then

$$(cf)(hgh^{-1}) = c \cdot f(hgh^{-1}) = c \cdot f(g) = (cf)(g),$$

so $cf \in R(G)$.

All three axioms hold, so $R(G)$ is a subspace of $\mathcal{F}_G$.

16. Use the character table of $\mathfrak{S}_4$ and Burnside's counting lemma to determine the number of distinct ways to color the faces of a cube with k colors, where two colorings are considered the same if one can be obtained from the other by a rotation.

Hint: The rotation group of the cube is isomorphic to $\mathfrak{S}_4$.

Solution:

Let G be the rotation group of the cube. By the hint, $G \cong \mathfrak{S}_4$ via the action of G on the four *long diagonals* of the cube; in particular $|G| = 24$. Let

$$X = \{\text{colorings } c: \{6 \text{ faces}\} \rightarrow \{1, \dots, k\}\}, \quad |X| = k^6,$$

on which G acts by $(g \cdot c)(f) = c(g^{-1}f)$. Two colorings are equivalent iff they lie in the same G -orbit, so the answer is $|X/G|$.

Step 1: Burnside via the character of the permutation representation.

Linearize the G -set X to get the permutation representation $V := \mathbb{C}X$ of G . By Maschke's theorem, V decomposes as a direct sum of irreducibles

$$V \cong_G \bigoplus_{\rho \in \text{Irr}(G)} m_\rho \cdot \rho, \quad m_\rho = \langle \chi^V, \chi^\rho \rangle.$$

The multiplicity of the trivial representation is, by Problem 9 and the orthogonality formula,

$$m_1 = \langle \chi^V, \chi^1 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi^V(g) = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}_X(g)|.$$

On the other hand, the component $V^G \subseteq V$ formed by the copies of the trivial representation, has a basis indexed by the G -orbits on X (the orbit sums $\sum_{x \in \mathcal{O}} x$, one per orbit $\mathcal{O} \in X/G$, are linearly independent and span V^G). Hence

$$|X/G| = \dim V^G = m_1 = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}_X(g)|.$$

This is the character-theoretic incarnation of Burnside's counting lemma.

Step 2: The character table of $\mathfrak{S}_4$. Recall that $\mathfrak{S}_4$ has 5 conjugacy classes, and hence 5 irreducible representations. The character table is:

$\mathfrak{S}_4$ class size	e	(ab)	$(ab)(cd)$	(abc)	$(abcd)$
1	1	1	1	1	1
sgn	1	-1	1	1	-1
W (2-dim)	2	0	2	-1	0
V_3 (standard, 3-dim)	3	1	-1	0	-1
$\text{sgn} \otimes V_3$	3	-1	-1	0	1

The leftmost row gives the conjugacy class data we need: 5 classes of sizes 1, 6, 3, 8, 6, summing to $24 = |\mathfrak{S}_4|$.

Step 3: Compute χ^V on each class. Under $G \cong \mathfrak{S}_4$, each conjugacy class corresponds to a geometric rotation type. By Problem 9, $\chi^V(g) = |\text{Fix}_X(g)| = k^{c(g)}$, where $c(g)$ counts the cycles of g 's action on the 6 faces. We tabulate:

$\mathfrak{S}_4$ class	size	rotation type	cycles on 6 faces	χ^V
e	1	identity	1+1+1+1+1+1	k^6
(ab)	6	180° through edge midpoints	2+2+2	k^3
$(ab)(cd)$	3	180° through face centers	1+1+2+2	k^4
(abc)	8	$\pm 120^\circ$ through vertex pair	3+3	k^2
$(abcd)$	6	$\pm 90^\circ$ through face centers	1+1+4	k^3

(For the 90° face rotation: the two faces on the rotation axis are fixed and the four side faces form a 4-cycle; for the 180° face rotation: the same two are fixed and the

four side faces split into two transpositions; for the 120° vertex rotation: the three faces meeting at one corner are cycled, and so are the three at the opposite corner; for the 180° edge rotation: the six faces split into three transpositions.)

Step 4: Apply the formula. By Step 1,

$$\begin{aligned} |X/G| &= \langle \chi^V, \chi^1 \rangle = \frac{1}{24} \sum_{\text{classes}} (\text{size}) \cdot \chi^V \cdot \overline{\chi^1} = \frac{1}{24} (1 \cdot k^6 + 6 \cdot k^3 + 3 \cdot k^4 + 8 \cdot k^2 + 6 \cdot k^3) \\ &= \boxed{\frac{1}{24} (k^6 + 3k^4 + 12k^3 + 8k^2)}. \end{aligned}$$

Sanity checks.

- $k = 1$: $\frac{1}{24}(1 + 3 + 12 + 8) = \frac{24}{24} = 1$ (the unique monochromatic pattern).
- $k = 2$: $\frac{1}{24}(64 + 48 + 96 + 32) = \frac{240}{24} = 10$.
- $k = 3$: $\frac{1}{24}(729 + 243 + 324 + 72) = \frac{1368}{24} = 57$.

Alternative solution (Burnside directly). One can avoid the language of representations and apply Burnside's counting lemma straight to the G -set X :

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}_X(g)|.$$

A coloring $c \in X$ is fixed by $g \in G$ iff c is constant on each cycle of g acting on the 6 faces, so $|\text{Fix}_X(g)| = k^{c(g)}$. Using the same table of conjugacy class sizes and cycle counts as in Step 3 above, we obtain the same total

$$|X/G| = \frac{1}{24} (k^6 + 3k^4 + 12k^3 + 8k^2).$$

This is essentially the same calculation: in this approach the conjugacy class data of $\mathfrak{S}_4$ is used implicitly (as the partition of G over which the sum is taken), whereas in the main solution above it appears explicitly through the orthogonality relation $|X/G| = \langle \chi^V, \chi^1 \rangle$.

Section 3: Test 2 Review

3.1 Theorems and Proofs of Theorems

Lemma 3.1 (Lemma 3.2). *If X is a G -set, then $G_{gx} = gG_xg^{-1}$ for any $g \in G$ and $x \in X$.*

Proof. An element $u \in G$ stabilizes gx if and only if $g^{-1}ug$ stabilizes x , which occurs if and only if $u \in gG_xg^{-1}$. $\square$

Proposition 3.2 (Proposition 3.4). *If X is a transitive G -set, then $X \cong G/G_x$ as G -sets for any $x \in X$.*

Proof. Let $x \in X$, and define $\varphi : G/G_x \rightarrow X$ by $\varphi(gG_x) = gx$ for $g \in G$.

If $gG_x = g'G_x$ for $g, g' \in G$, then $g^{-1}g' \in G_x$ and hence $g^{-1}g'x = x$, giving $gx = g'x$. This shows that φ is well-defined. By reversing the argument, we see that φ is injective.

For any $u \in G$ and $gG_x \in G/G_x$, we have

$$u\varphi(gG_x) = u(gx) = (ug)x = \varphi(ugG_x) = \varphi(u(gG_x)),$$

showing that φ is a G -set homomorphism.

For any $y \in X$, by transitivity there exists some $g \in G$ such that $y = gx = \varphi(gG_x)$, so φ is surjective.

Therefore, φ is a G -set isomorphism. $\square$

Lemma 3.3 (Lemma 3.6). *Let $\varphi : X \rightarrow Y$ be a homomorphism of G -sets, and let $x \in X$. Then $G_x \leq G_{\varphi(x)}$, and if φ is an isomorphism, then $G_x = G_{\varphi(x)}$.*

Proof. If $g \in G_x$, then $\varphi(x) = \varphi(gx) = g\varphi(x)$; it follows that $g \in G_{\varphi(x)}$. Hence $G_x \leq G_{\varphi(x)}$.

If φ is an isomorphism, then by considering the G -set homomorphism $\varphi^{-1} : Y \rightarrow X$, we have

$$G_{\varphi(x)} \leq G_{\varphi^{-1}(\varphi(x))} = G_x.$$

Therefore, $G_x = G_{\varphi(x)}$. $\square$

Corollary 3.4 (Corollary 3.5). *Let X be a G -set. Then $Gx \cong G/G_x$ as G -sets for any $x \in X$. In particular, if G is finite, then $|Gx| = [G : G_x]$.*

Proposition 3.5 (Proposition 3.3). *Any G -set has a unique partition consisting of transitive G -sets, namely its partition into orbits.*

Proof. First, note that if X has a partition consisting of transitive G -sets, then those sets must be the orbits, since the orbits are precisely the transitive subsets of X . This proves uniqueness.

For existence, let $x \in X$. Then x lies in the orbit Gx , so it suffices to show that any two orbits of X under the action of G are either equal or disjoint.

Let $x, y \in X$ and suppose that $Gx \cap Gy$ is non-empty. Then there exist $g_1, g_2 \in G$ such that

$$g_1x = g_2y.$$

It follows that

$$y = g_2^{-1}g_1x \in Gx,$$

and hence $Gy \subseteq Gx$. By symmetry, we also have $Gx \subseteq Gy$, and therefore $Gx = Gy$. $\square$

Theorem 3.6 (Cayley's Theorem). *Every group G is isomorphic to a subgroup of $\text{Sym}(G)$. In particular, if G is finite with $|G| = n$, then G is isomorphic to a subgroup of $\text{Sym}(n)$.*

Proof. The group G acts on itself by left multiplication, i.e., for $g, x \in G$, define $g \cdot x = gx$. This is the case $H = \{1\}$ of the action of G on the coset space G/H .

If $g \in G$ satisfies $gx = x$ for all $x \in G$, then taking $x = g^{-1}$ gives $g^{-1} = gg^{-1} = 1$, and hence $g = 1$. Therefore, the action is faithful.

Thus, we obtain an injective homomorphism (a monomorphism) from G into $\text{Sym}(G)$, and the result follows. $\square$

Theorem 3.7 (Sylow's Theorem). *Let G be a finite group and let p be a prime dividing $|G|$. Then the following hold:*

1. G has at least one Sylow p -subgroup.
2. All Sylow p -subgroups of G are conjugate.
3. Any p -subgroup of G is contained in a Sylow p -subgroup.
4. The number of Sylow p -subgroups of G is congruent to 1 (mod p).

Theorem 3.8 (Sylow's Theorem's Alternative). *Let p be prime and G a group of order $|G| = p^a m$, and $\gcd(p^a, m) = 1$. Then*

- (1) Every p -subgroup is contained in a Sylow p -subgroup. In particular, Sylow subgroups exist.
- (2) If n_p is the number of Sylow p -subgroups, then $n_p \equiv 1 \pmod{p}$
- (3) $n_p | m$
- (4) all Sylow p -subgroups are conjugate of each other

Theorem 3.9 (Cauchy's Theorem). *Let G be a finite group and let p be a prime dividing $|G|$. Then G has an element of order p .*

Proof. By part (i) of Sylow's Theorem, G has a nontrivial Sylow p -subgroup. Hence G contains a nontrivial p -element, some power of which has order p . $\square$

Theorem 3.10 ($p < q$ Theorem). *Let p, q be two primes with $p < q$ and $|G| = pq$. Then G has only one q -subgroup and so this q -subgroup is normal. Additionally, whenever $q \not\equiv 1 \pmod{p}$, then there is a unique group of order pq which is $\mathbb{Z}_{pq}$*

Proof. From Sylow's theorem we know that $n_q | p$, and $n_q \equiv 1 \pmod{p}$. This tells us that $n_q = 1, 1+p, 1+2p, \dots$. However, $1+kp > p$ for $k > 0$. So it must be the case the $n_q = 1$. Let Q be this Sylow q -subgroup. This means that $Q \triangleleft G$. Similarly, from Sylow's theorem, we know that $n_p | q$, and $n_p \equiv 1 \pmod{p}$. This means $n_p = 1, 1+p, 1+2p, \dots$. If $q \not\equiv 1 \pmod{p}$, then $n_p = 1$. Let P be this Sylow p -subgroup. Since $n_p = 1$,

we know that $P \triangleleft G$. We know that $P \cap Q = \{e\}$ since each group is of prime order; meaning they are both cyclic and generated by differing elements. Now that we know both P and Q are normal, we see that the internal direct product $PQ = QP \leq G$. The statement $P \cap Q = \{e\}$ and $PQ = QP \leq G$ show that $|PQ| = |QP| = pq = |G|$. Hence $PQ = G$. Furthermore, we see that $G = PQ \cong P \times Q$. Since both P and Q are cyclic groups of prime order $P \cong \mathbb{Z}_p$ and $Q \cong \mathbb{Z}_q$, thus $G \cong P \times Q \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq}$. $\square$

Theorem 3.11 (Theorem 8.1). *If P is a nontrivial finite p -group, then $Z(P)$ is also nontrivial. Moreover, if $N \neq \{1\}$ is a normal subgroup of P , then $N \cap Z(P) \neq \{1\}$.*

Proof. Suppose that $Z(P) = \{1\}$. Let $K_1 = \{1\}, K_2, \dots, K_r$ be the conjugacy classes of P . If $K_i = \{x\}$ for some i and some $x \in P$, then $gxg^{-1} = x$ for all $g \in P$, which shows that $x \in Z(P) = \{1\}$. Therefore, $|K_i| > 1$ for each $1 < i \leq r$.

Each K_i is an orbit under the action of P on itself by conjugation, so $|K_i|$ divides $|P|$ for all i . Since P is a p -group, it follows that $|K_i| \equiv 0 \pmod{p}$ for each $1 < i \leq r$.

Now,

$$|P| = |K_1| + |K_2| + \dots + |K_r| = 1 + |K_2| + \dots + |K_r| \equiv 1 \pmod{p},$$

since P is the disjoint union of $K_1, \dots, K_r$. This is a contradiction, since $|P|$ is divisible by p . Therefore, $Z(P)$ is nontrivial.

Now let $N \leq P$ with $N \neq \{1\}$. Then N is a union of some of the conjugacy classes K_j , where the index set $S \subseteq \{1, \dots, r\}$ contains 1 and $|S| > 1$. If $N \cap Z(P) = \{1\}$, then $|K_j| > 1$ for all $j \in S$ with $j \neq 1$, and as above this would imply

$$|N| \equiv 1 \pmod{p},$$

which is impossible since $|N|$ is divisible by p . Therefore, $N \cap Z(P) \neq \{1\}$. $\square$

Lemma 3.12 (Lemma 8.2). *If G is a non-abelian group, then $G/Z(G)$ cannot be cyclic. In particular, if P is a finite p -group, then $[P : Z(P)] \neq p$.*

Proof. Suppose that $G/Z(G) = \langle xZ(G) \rangle$ for some $x \in G$. Then G is generated by the set $S = Z(G) \cup \{x\}$. Every pair of elements in S commutes: elements of $Z(G)$ commute with all elements of G , and hence with x , while elements of $Z(G)$ commute with each other. Therefore, all elements of S commute, and it follows that $\langle S \rangle = G$ is abelian, a contradiction.

For the second statement, if P is a finite p -group and $[P : Z(P)] = p$, then $P/Z(P)$ has order p and is therefore cyclic, contradicting the first part. $\square$

Theorem 3.13 (Theorem 8.7). *The following statements about a finite group G are equivalent:*

1. *Every Sylow subgroup of G is normal in G .*
2. *G is the direct product of its Sylow subgroups.*
3. *Every maximal subgroup of G is normal in G .*

Proposition 3.14 (Proposition 10.1). *Finite groups have composition series.*

Proof. Let G be a finite group. We proceed by induction on $|G|$.

If G is simple, then $G > \{1\}$ is a composition series of G .

Otherwise, G has a maximal normal subgroup G_1 , which by induction has a composition series

$$G_1 > G_2 > \cdots > G_r = \{1\}.$$

Since G/G_1 is simple, it follows that

$$G = G_0 > G_1 > \cdots > G_r = \{1\}$$

is a composition series of G . □

Lemma 3.15 (Lemma 10.2). *Let G be a group that has a composition series, and let $N \trianglelefteq G$. Then N has a composition series.*

Theorem 3.16 (Jordan–Hölder Theorem). *Suppose that G is a group that has a composition series. Then any two composition series of G have the same length and are equivalent.*

3.2 Important Definitions for Test 2

Definition 3.1. Let G be a group $|G| = p^a m$, and $\gcd(p^a, m) = 1$. A p -subgroup is a subgroup $H \leq G$ with $|H| = p^k$ for $k \leq a$. If $k = a$ we call it a Sylow p -subgroup.

Theorem 3.17 (The Structure of Finite Abelian p -Groups). *Every finite abelian p -group is isomorphic to a direct product*

$$G \cong \mathbb{Z}_{p^{a_1}} \times \cdots \times \mathbb{Z}_{p^{a_k}} \quad (3.1)$$

Theorem 3.18 (Fundamental Theorem of Finite Abelian Groups). *Every finite abelian group is uniquely isomorphic to $G = \mathbb{Z}_{q_1^{a_1}} \times \cdots \times \mathbb{Z}_{q_r^{a_r}}$. Where q_r are (not necessarily distinct) primes.*

Lemma 3.19. *Every finite abelian group is isomorphic to the product of its Sylow subgroups, that is if $|G| = p_1^{a_1} \cdots p_k^{a_k}$ and P_i is the unique Sylow P_i -subgroup, then $G \cong P_1 \times P_2 \times \cdots \times P_k$*

Definition 3.2 (Subnormal Series). A subnormal series of G is a chain

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq G_3 \trianglelefteq \cdots \trianglelefteq G_n = G \quad (3.2)$$

Definition 3.3 (The Factors of A Series). The quotients G_{i+1}/G_i are the factors of the series.

Definition 3.4 (A Composition Series of G). A composition series of G is a subnormal series:

$$\{e\} \triangleleft G_1 \triangleleft G_2 \triangleleft \cdots \triangleleft G_n = G \quad (3.3)$$

here G_{i+1}/G_i is a *simple group*. This means that G_{i+1}/G_i has no proper normal subgroups.

Example 3.1. Let $\mathbb{Z}/12\mathbb{Z}$ We can see the composition series of this becomes

$$\{0\} \triangleleft \langle 6 \rangle \triangleleft \langle 3 \rangle \triangleleft \langle 1 \rangle = \mathbb{Z}/12\mathbb{Z} \quad (3.4)$$

We can look at some of the factors of the composition series

$$\langle 6 \rangle / \langle 0 \rangle \cong \mathbb{Z}/2\mathbb{Z} \quad (3.5)$$

$$\langle 3 \rangle / \langle 6 \rangle \cong \mathbb{Z}/2\mathbb{Z} \quad (3.6)$$

$$\langle 1 \rangle / \langle 3 \rangle \cong \mathbb{Z}/3\mathbb{Z} \quad (3.7)$$

3.3 Alternative Answers For Test 2

Problem 3.20 (QUESTION 1). *This problem is divided into several parts:*

(a) *Let X and Y be G -sets. A G -set **isomorphism** is a bijection $\phi : X \rightarrow Y$ such that*

$$\phi(g \cdot x) = g \cdot \phi(x) \quad (3.8)$$

for all $g \in G$ and $x \in X$.

(b)—(his solution is good)—

(c)—(his solution is good)—

(d)—(his solution is good)—

(e)—(his solution is good)—

(f)—(his solution is good)—

(g)—(his solution is good)—

Problem 3.21 (QUESTION 2). *An element $u \in G$ stabilizes gx if and only if $g^{-1}ug$ stabilizes x , which occurs if and only if $u \in gG_xg^{-1}$.*

Problem 3.22 (QUESTION 3). *Let $x \in X$, and define $\varphi : G/G_x \rightarrow X$ by $\varphi(gG_x) = gx$ for $g \in G$.*

If $gG_x = g'G_x$ for $g, g' \in G$, then $g^{-1}g' \in G_x$ and hence $g^{-1}g'x = x$, giving $gx = g'x$. This shows that φ is well-defined. By reversing the argument, we see that φ is injective.

For any $u \in G$ and $gG_x \in G/G_x$, we have

$$u\varphi(gG_x) = u(gx) = (ug)x = \varphi(ugG_x) = \varphi(u(gG_x)),$$

showing that φ is a G -set homomorphism.

For any $y \in X$, by transitivity there exists some $g \in G$ such that $y = gx = \varphi(gG_x)$, so φ is surjective.

Therefore, φ is a G -set isomorphism.

Problem 3.23 (QUESTION 4). *Recall that if we let G be a group $|G| = p^am$, and $\gcd(p^a, m) = 1$. A p -subgroup is a subgroup $H \leq G$ with $|H| = p^k$ for $k \leq a$. If $k = a$ we call it a Sylow p -subgroup. Our goal is to show that if for $k = a$, $H \trianglelefteq G$, and $|P| = p^a$, then $H \subseteq P$. We can start off by seeing H as a p -group where $k = 1$, since we let $|H| = p$, from Sylow's theorem we know that $H \subseteq P'$ for some Sylow p -subgroup P' . Since $H \trianglelefteq G$, $gHg^{-1} = H$, $H = gHg^{-1} \subseteq gP'g^{-1}$ for some $g \in G$. However, because all Sylow p -subgroups are conjugate of one another, $gP'g^{-1} = P$, where P is any arbitrary Sylow p -subgroup. So $H \subseteq P$.*

Problem 3.24 (QUESTION 5). *We will begin by decomposing the order of G . We can see that $|G| = 5 \cdot 8 = 40$. From Sylow's theorem, we know that the number of Sylow 5-subgroups, $n_5 \equiv 1 \pmod{5}$, and that $n_5 | 8$. With this in mind the only divisors of 8 that are congruent to 1, so $n_5 = 1$. Since the subgroup is unique it must be normal. This tells us that all groups of order 40 are not simple as they have at least one normal subgroup of order 5.*

Problem 3.25 (QUESTION 6). *Base Case: If $|G| = 2$, then $G \cong \mathbb{Z}/2\mathbb{Z}$ is simple, and $\{e\} \triangleleft G$ is a composition series.*

Inductive Step: Assume that every group of order less than $|G|$ has a composition series.

- *If G is a simple group, then $\{e\} \triangleleft G$ is a composition series (we are done).*

- If G is not simple, then G has a proper nontrivial normal subgroup. Among all of those subgroups, choose N to be the maximal proper normal subgroup of G . From this we know that G/N is simple because Any normal subgroup of G/N would correspond to a normal subgroup of G containing N . By maximality, this must be G/N itself or it is trivial. Since $|N| < |G|$, by the induction hypothesis, N has a composition series:

$$\{e\} = N_0 \triangleleft N_1 \triangleleft \cdots \triangleleft N_k = N \quad (3.9)$$

Then:

$$\{e\} = N_0 \triangleleft N_1 \triangleleft \cdots \triangleleft N_k = N \triangleleft G \quad (3.10)$$

Because every factor N_{i+1}/N_i is simple and G/N is simple

Problem 3.26 (QUESTION 7). If we look at the conjugation action $G^{\circ G}$ with $\text{Fix}(G) = Z(G)$, by the class equation we know

$$|G| = |\text{Fix}(G)| + \sum_{h \in G} [G : G_h]. \quad (3.11)$$

We also know that $|G|$ needs to be divided by each $[G : G_h]$. So we must have $[G : G_h] \mid |G| = p^a$; hence, we require $p \mid [G : G_h]$. So our class equation is now

$$|G| = p^a = |Z(G)| + p^{b_1} + \cdots + p^{b_n}, \quad (3.12)$$

where $b_i < a$. So it must be the case that $p \mid |Z(G)|$, meaning $Z(G)$ is nontrivial.

Problem 3.27 (QUESTION 8). Suppose we let G be an abelian group of order m and have $n \mid m$. We will show that G has a subgroup of order n . We can consider the prime factorizations of n and m :

$$\prod_{i=1}^r p_i^{a_i} = m, \quad \prod_{i=1}^r p_i^{b_i} = n \quad (3.13)$$

where $0 \leq b_i \leq a_i$ for $i = 1, 2, \dots, r$. Since we know that G is a finite abelian group, it is the direct product of Sylow p_i -subgroups:

$$G \cong G_{p_1} \times G_{p_2} \times \cdots \times G_{p_r} \quad (3.14)$$

where each G_{p_i} is an abelian p_i -group of order $p_i^{a_i}$. So it is enough to show that if P is a finite abelian group of order p^a , then for any $0 \leq b \leq a$, there exists a subgroup H of order b in P . This is true since we can represent a subgroup of order n of G by a direct product of p_i -subgroups that match the structure of the prime factorization of n . For example if $H_i \leq G_{p_i}$ where $|H_i| = p_i^{b_i}$, then

$$H_1 \times H_2 \times \cdots \times H_r \leq G \quad (3.15)$$

has order $\prod_{i=1}^r p_i^{b_i} = n$. Now if we let P be a finite abelian group of order p^a , the fundamental theorem of finite abelian groups tells us that

$$P \cong \mathbb{Z}_{p^{e_1}} \times \mathbb{Z}_{p^{e_2}} \times \cdots \times \mathbb{Z}_{p^{e_t}} \quad (3.16)$$

where $e_1 + e_2 + \cdots + e_t = a$. For each of the $\mathbb{Z}_{p^{e_i}}$ there is a subgroup of order $\mathbb{Z}_{p^{c_i}}$. For each of these groups in the direct product we can adjust each c_i to get $c_1 + c_2 + \cdots + c_t = b$. The corresponding direct product of subgroups will be of order $p^{c_1} \cdot p^{c_2} \cdots p^{c_t} = p^b$. Hence P has a subgroup of order p^b .

3.4 Test 2 Solutions

NAME(print): _____

Note: There is a total of 49 points available in this test. The test is graded out of 37 points and any number of points (up to 12) exceeding this value will count as extra credit for Midterm 1.

1. (14 points) Answer each of the following questions precisely but briefly (no need of a proof).

(a) (2 points) Give the definition of an isomorphism of G -sets.

Solution: Let X and Y be G -sets. An *isomorphism of G -sets* is a bijection $\varphi : X \rightarrow Y$ such that $\varphi(g \cdot x) = g \cdot \varphi(x)$ for all $g \in G$ and $x \in X$.

(b) (2 points) Give the definition of a transitive G -set.

Solution: A G -set X is called *transitive* if for every pair of elements $x, y \in X$, there exists some $g \in G$ such that $g \cdot x = y$. Equivalently, X has exactly one orbit under the action of G .

(c) (2 points) Give a representative for each of the **distinct** irreducible permutation representations of A_4 . How many of these are there in total?

Solution: The distinct irreducible permutation representations of A_4 correspond to the conjugacy classes of subgroups of A_4 . The subgroups of A_4 are:

- $\{e\}$ (trivial subgroup), giving the regular representation on 12 elements.
- $\langle (12)(34) \rangle \cong \mathbb{Z}_2$, giving a transitive representation on $[A_4 : \mathbb{Z}_2] = 6$ elements. (The three subgroups of order 2 are all conjugate in A_4 .)
- $\langle (123) \rangle \cong \mathbb{Z}_3$, giving a transitive representation on $[A_4 : \mathbb{Z}_3] = 4$ elements. (The four subgroups of order 3 are all conjugate in A_4 .)
- $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$, giving a transitive representation on $[A_4 : V_4] = 3$ elements.
- A_4 itself, giving the trivial representation on 1 element.

There are **5** distinct irreducible permutation representations in total.

- (d) (2 points) Give the definition of a Sylow p -subgroup of a finite group G .

Solution: Let G be a finite group and let p be a prime. Write $|G| = p^a m$ where $a \geq 1$ and $\gcd(p, m) = 1$. A *Sylow p -subgroup* of G is a subgroup $P \leq G$ of order p^a , i.e., a p -subgroup of G of the largest possible order.

- (e) (2 points) Give the definition of a linear representation of a group G .

Solution: A *linear representation* of a group G is a group homomorphism $\rho : G \rightarrow GL(V)$, where V is a vector space over a field k and $GL(V)$ denotes the group of invertible linear transformations from V to itself.

- (f) (2 points) What are the possible abelian groups of order 36 up to isomorphism?

Solution: By the Fundamental Theorem of Finite Abelian Groups, we factor $36 = 2^2 \cdot 3^2$ and list all possible abelian groups as products of cyclic groups corresponding to partitions of each prime-power factor:

- $\mathbb{Z}_4 \times \mathbb{Z}_9 \cong \mathbb{Z}_{36}$
- $\mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3$

There are 4 abelian groups of order 36 up to isomorphism.

- (g) (2 points) Give a composition series for $\mathbb{Z}_{12}$. What are the composition factors of this series?

Solution: A composition series for $\mathbb{Z}_{12}$ is:

$$\{0\} \trianglelefteq \langle 6 \rangle \trianglelefteq \langle 3 \rangle \trianglelefteq \langle 1 \rangle = \mathbb{Z}_{12},$$

i.e., $\{0\} \trianglelefteq \mathbb{Z}_2 \trianglelefteq \mathbb{Z}_4 \trianglelefteq \mathbb{Z}_{12}$, where we identify $\langle 6 \rangle \cong \mathbb{Z}_2$, $\langle 3 \rangle \cong \mathbb{Z}_4$. The composition factors (successive quotients) are:

- $\langle 6 \rangle / \{0\} \cong \mathbb{Z}_2$,
- $\langle 3 \rangle / \langle 6 \rangle \cong \mathbb{Z}_2$,
- $\mathbb{Z}_{12} / \langle 3 \rangle \cong \mathbb{Z}_3$.

So the composition factors are $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3$.

2. (5 points) If X is a G -set, prove that $G_{g \cdot x} = gG_xg^{-1}$ for any $g \in G$ and $x \in X$.

Solution: Recall that for any $y \in X$, the stabilizer is $G_y = \{h \in G : h \cdot y = y\}$.

($\subseteq$) Let $h \in G_{g \cdot x}$, so $h \cdot (g \cdot x) = g \cdot x$. Applying g^{-1} on the left, we get $g^{-1}hg \cdot x = x$, which means $g^{-1}hg \in G_x$. Therefore $h = g(g^{-1}hg)g^{-1} \in gG_xg^{-1}$.

($\supseteq$) Let $h \in gG_xg^{-1}$, so $h = gag^{-1}$ for some $a \in G_x$ (meaning $a \cdot x = x$). Then

$$h \cdot (g \cdot x) = (gag^{-1}) \cdot (g \cdot x) = g \cdot (a \cdot (g^{-1}g \cdot x)) = g \cdot (a \cdot x) = g \cdot x.$$

Hence $h \in G_{g \cdot x}$.

We conclude that $G_{g \cdot x} = gG_xg^{-1}$.

3. (5 points) Prove that if X is a transitive G -set, then X is isomorphic to the G -set G/G_x of left cosets of G_x in G , where $x \in X$ is any element.

Solution: Fix $x \in X$ and define $\varphi : G/G_x \rightarrow X$ by $\varphi(gG_x) = g \cdot x$.

Well-defined: If $gG_x = hG_x$, then $h^{-1}g \in G_x$, so $h^{-1}g \cdot x = x$, which gives $g \cdot x = h \cdot x$. Hence φ is well-defined.

G-equivariant: For any $g, h \in G$, we have $\varphi(g \cdot hG_x) = \varphi((gh)G_x) = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot \varphi(hG_x)$.

Surjective: Let $y \in X$. Since X is transitive, there exists $g \in G$ with $g \cdot x = y$. Then $\varphi(gG_x) = y$.

Injective: If $\varphi(gG_x) = \varphi(hG_x)$, then $g \cdot x = h \cdot x$, so $h^{-1}g \cdot x = x$, meaning $h^{-1}g \in G_x$, and thus $gG_x = hG_x$.

Therefore φ is a G -set isomorphism, and $X \cong G/G_x$.

4. (5 points) Let H be a normal subgroup of a finite group G and assume that $|H| = p$, where p is a prime. Show that H is contained in every p -Sylow subgroup of G .

Solution: Since $|H| = p$, the subgroup H is a p -subgroup of G . By Sylow's theorem, every p -subgroup of G is contained in some Sylow p -subgroup, so there exists a Sylow p -subgroup Q of G with $H \leq Q$. Now let P be any Sylow p -subgroup of G . By Sylow's theorem, all Sylow p -subgroups are conjugate, so $P = gQg^{-1}$ for some $g \in G$. Since H is normal in G , we have $gHg^{-1} = H$, and therefore

$$H = gHg^{-1} \leq gQg^{-1} = P.$$

Since P was an arbitrary Sylow p -subgroup, H is contained in every Sylow p -subgroup of G .

5. (5 points) Prove that there are no simple groups of order 40.

Solution: Let G be a group of order $40 = 2^3 \cdot 5$. By Sylow's theorem, the number n_5 of Sylow 5-subgroups satisfies $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 8$. The divisors of 8 are 1, 2, 4, 8, and among these only $n_5 = 1$ satisfies $n_5 \equiv 1 \pmod{5}$.

Therefore there is a unique Sylow 5-subgroup P of G . Since P is the unique Sylow 5-subgroup, it must be normal in G (as gPg^{-1} is also a Sylow 5-subgroup for any $g \in G$, so $gPg^{-1} = P$). Since P is a proper nontrivial normal subgroup ($|P| = 5$, and $1 < 5 < 40$), G is not simple.

6. (5 points) Prove that every finite group has a composition series.

Solution: We proceed by strong induction on $|G|$.

Base case: If $|G| = 1$, then the trivial series $\{e\}$ is a composition series (it has no factors to check).

Inductive step: Assume every group of order less than $|G|$ has a composition series. If G is simple, then $\{e\} \trianglelefteq G$ is a composition series (the only factor is $G/\{e\} \cong G$, which is simple). If G is not simple, then G has a proper nontrivial normal subgroup. Among all proper nontrivial normal subgroups of G , choose a maximal one, say N (this exists since G is finite). Then G/N is simple: if G/N had a proper nontrivial normal subgroup $\bar{M}$, by the correspondence theorem $\bar{M} = M/N$ for some normal subgroup M of G with $N \subsetneq M \subsetneq G$, contradicting the maximality of N .

Since $|N| < |G|$, by the induction hypothesis N has a composition series

$$\{e\} = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_k = N.$$

Appending G at the top gives the series

$$\{e\} = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_k = N \trianglelefteq G,$$

where each factor N_{i+1}/N_i is simple (by the composition series of N) and G/N is simple by construction. This is a composition series for G .

7. (5 points) Prove that in a non-trivial finite p -group G , the center $Z(G)$ is non-trivial.

Solution: Let G be a finite p -group with $|G| = p^n$ for some $n \geq 1$. Consider the action of G on itself by conjugation, i.e., $g \cdot x = gxg^{-1}$. The orbits of this action are the conjugacy classes of G , and the class equation gives

$$|G| = |Z(G)| + \sum_i |G : G_{x_i}|,$$

where the sum runs over one representative x_i from each conjugacy class of size greater than 1, and G_{x_i} is the centralizer of x_i .

For each non-central element x_i , the orbit size $|G : G_{x_i}|$ divides $|G| = p^n$ and is greater than 1, so $p \mid |G : G_{x_i}|$. Since $p \mid |G|$ and p divides each term in the sum, it follows that $p \mid |Z(G)|$. In particular, $|Z(G)| \geq p > 1$, so $Z(G)$ is non-trivial.

8. (5 points) Let G be an abelian group of order m . If n divides m , prove that G has a subgroup of order n .

Solution: By the Fundamental Theorem of Finitely Generated Abelian Groups, we may write

$$G \cong \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}},$$

where $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ (not necessarily distinct primes). Since $n \mid m$, we can write $n = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$ where $0 \leq b_i \leq a_i$ for each i .

In each cyclic factor $\mathbb{Z}_{p_i^{a_i}}$, there is a unique subgroup of order $p_i^{b_i}$, namely the subgroup generated by $p_i^{a_i-b_i}$ (since cyclic groups have exactly one subgroup of each order dividing the group order). Call this subgroup $H_i \leq \mathbb{Z}_{p_i^{a_i}}$.

Then

$$H = H_1 \times H_2 \times \cdots \times H_k \leq \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}} \cong G$$

is a subgroup of order $|H| = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k} = n$.

Section 4: Test 1 Review

4.1 Alternative Answers for Test 1

Problem 4.1 (QUESTION 1). *The answers are enumerated as follows:*

- (a) *Let $f : H \rightarrow G$ be a homomorphism. Then we know that $\text{Ker } f \triangleleft H$, $\text{Im } f \leq G$, and $H/\text{Ker } f \cong \text{Im } f$*
- (b) *—(His Solution is Good)—*
- (c) *—(His Solution is Good)—*
- (d) *—(His Solution is Good)—*
- (e) *—(His Solution is Good)—*
- (f) *—(His Solution is Good)—*

Problem 4.2 (QUESTION 2). *Since $\mathcal{C}$ is a family of subsets of G that form a partition of G , we know $\exists C \in \mathcal{C}$ where $e \in C$. We will show that this particular C is a subgroup of G . Suppose G is of the form $G = \{e, x_1, x_2, \dots, x_{n-2}, x_{n-1}\}$ where n is the order of the group.*

1. **Existence of Identity:** *By construction we know that $e \in C$.*
2. **Closure Under Inverses:** *Take any $x_i \in C$. If we observe $x_i C$ we see that $x_i \in x_i C$ since $e \in C$ and $x_i = x_i e \in x_i C$. However, since $x_1 C \in \mathcal{C}$, meaning $x_i C$ is a partition, and $x_i \in x_i C$, it must be the case that $x_i C = C$. From this we know that $e \in x_i C$ since $x_i C = C$, or, in other words, $e = x_i x_j$ for some $x_j \in C$. The only element in G that can satisfy this is x_i^{-1} . Here we can conclude that $x_i^{-1} \in C$.*
3. **Closure Under Operations:** *Take $x_i, x_j \in C$. We see that $x_j x_i \in x_j C$. However, we showed for any $x_j \in C$, $x_j C = C$. So $x_j x_i \in C = x_j C$.*

Now we will show that $\mathcal{C}$ is a set of cosets of C in G , or $\mathcal{C} = G/C$. We will first observe the definition of the set G/C

$$G/C = \{gC | g \in G\} \quad (4.1)$$

From the premise, we know that $gC \in \mathcal{C} \quad \forall g \in G$. With this and the definition of G/C , we see that $G/C \subseteq \mathcal{C}$. Furthermore, we can take any random partition $D \in \mathcal{C}$. If we take some element $x_k \in D$, we see $e \in x_k^{-1} D$ since $e = x_k^{-1} x_k \in x_k^{-1} D$. This means $x_k^{-1} D = C$. If we left multiply by x_k we get $D = x_k x_k^{-1} D = x_k C$. Since $x_k \in G$, $D = x_k C \in G/C$, or $\mathcal{C} \subseteq G/C$

Problem 4.3 (QUESTION 3). *We take $\sigma \in \text{Aut}(G)$ and $\varphi_g \in \text{Inn}(G)$ for some $g \in G$. We observe if we have $\sigma\varphi_g\sigma^{-1}(x)$ for $x \in G$*

$$\sigma\varphi_g\sigma^{-1}(x) = \sigma(g\sigma^{-1}(x)g^{-1}) = \sigma(g)\sigma(\sigma^{-1}(x))\sigma^{-1}(g) = \sigma(g)x\sigma^{-1}(g) = \varphi_{\sigma(g)}(x). \quad (4.2)$$

Hence $\varphi_g(x) \in \text{Inn}(G)$, implying that $\sigma\text{Inn}(G)\sigma^{-1} \subseteq \text{Inn}(G)$. This shows that $\text{Inn}(G) \triangleleft \text{Aut}(G)$.

Problem 4.4 (QUESTION 4). *Let $\sigma : G \rightarrow \mathfrak{S}_{G/H} \cong \mathfrak{S}_p$ defined by $g \mapsto \sigma(g)(x) = gxH, x \in G$. This means that σ is a homomorphism. From this, we can see that:*

$$\text{Ker}(\sigma) = \{g \in G | gxH = xH, \forall x \in G\} \quad (4.3)$$

$$= \{g \in G | x^{-1}gxH = H, \forall x \in G\} \quad (4.4)$$

$$= \{g \in G | x^{-1}gx \in H, \forall x \in G\} \quad (4.5)$$

$$= \{g \in G | g \in x^{-1}Hx, \forall x \in G\} \quad (4.6)$$

$$= \bigcap_{x \in G} x^{-1}Hx \leq H, \quad (4.7)$$

since the intersection of subgroups is always a subgroup. Furthermore, by the first isomorphism theorem we know that $G/\text{Ker}(\sigma) \cong \text{Im}(\sigma) \implies |G/\text{Ker}(\sigma)| = |\text{Im}(\sigma)| |\mathfrak{S}_p| = p!$. But we know that $|G/\text{Ker}(\sigma)| = |G/H| \cdot |H/\text{Ker}(\sigma)| = p \cdot |K/\text{Ker}(\sigma)|$. From here we can observe that $p \cdot |K/\text{Ker}(\sigma)| p! = p \cdot (p-1)! \implies |K/\text{Ker}(\sigma)| (p-1)!$. We also see that since $\text{Ker}(\sigma) \leq H$, this implies that $|H/\text{Ker}(\sigma)| |H|$, but all primes that divide H are greater than or equal to p since $\frac{|G|}{|H|} = p$. So all primes that divide $|H/\text{Ker}(\sigma)|$ are greater than p . Since $|H/\text{Ker}(\sigma)| (p-1)!$, this means that $|H/\text{Ker}(\sigma)| = 1$. This implies that $H = \text{Ker}(\sigma) \triangleleft G$ since the kernel is always normal in G for a homomorphism.

Problem 4.5 (QUESTION 5). We must show that $G \cong H \leq \mathfrak{S}_G$. Let $\sigma : G \rightarrow \mathfrak{S}_G$ be a left permutation representation defined by $g \mapsto \sigma(g)(x) = g \cdot x$ for $x, g \in G$. From this, σ is readily seen as a homomorphism

$$\sigma(gh)(x) = gh \cdot x = g \cdot (h \cdot x) = \sigma(g)(\sigma(h)(x)) = \sigma(g) \circ \sigma(h)(x). \quad (4.8)$$

By the first isomorphism theorem, we know that $G/\text{Ker}(\sigma) \cong \text{Im}(\sigma)$. Since σ is a homomorphism, $\text{Ker}(\sigma) = \{e\}$ and $\text{Im}(\sigma) \leq \mathfrak{S}_G$. So we can see $G \cong G/\{e\} \cong \text{Im}(\sigma) \leq \mathfrak{S}_G$

Problem 4.6 (QUESTION 6). —(HIS SOLUTION IS GOOD)—

4.2 Test 1 Solutions

NAME(print):_____ ID _____

1. (12 points) Answer each of the following questions precisely but shortly (no need of a proof).
- (a) (2 points) State the First Isomorphism Theorem.

Solution: Let $\phi : G \rightarrow H$ be a group homomorphism. Then $\ker(\phi)$ is a normal subgroup of G , and the map $\bar{\phi} : G/\ker(\phi) \rightarrow \text{im}(\phi)$ defined by $\bar{\phi}(g\ker(\phi)) = \phi(g)$ is a well-defined group isomorphism. That is, $G/\ker(\phi) \cong \text{im}(\phi)$.

- (b) (2 points) Recall that all subgroups $H \leq \mathbb{Z}$ are all of the form $H = n\mathbb{Z}$ for some nonnegative integer n . Then the subgroup $n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$ for some integer d . What is d ?

Solution: The subgroup $n\mathbb{Z} + m\mathbb{Z}$ is the set of all integer linear combinations of n and m , i.e., $\{an + bm : a, b \in \mathbb{Z}\}$. This is precisely the subgroup generated by $\gcd(n, m)$, so $d = \gcd(n, m)$.

- (c) (2 points) What is the group $\text{Aut}(\mathbb{Z}_p)$ when p is prime?

Solution: It is isomorphic to $(\mathbb{Z}_p)^\times$, the group of units in $\mathbb{Z}_p$, which is a cyclic group of order $p - 1$. In particular, $\text{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$.

- (d) (2 points) Give the definition of group action.

Solution: A *group action* of a group G on a set X is a map $G \times X \rightarrow X$, denoted $(g, x) \mapsto g \cdot x$, satisfying:

- (i) (Identity) For all $x \in X$, $e \cdot x = x$ where e is the identity element of G .
- (ii) (Compatibility) For all $g, h \in G$ and $x \in X$, $(gh) \cdot x = g \cdot (h \cdot x)$.

- (e) (2 points) State Burnside's lemma.

Solution: Let G be a finite group acting on a finite set X . For each $g \in G$, let $\text{Fix}(g) = \{x \in X : g \cdot x = x\}$ be the set of elements of X fixed by g . Then the number k of orbits of G on X is given by

$$k = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|.$$

- (f) (2 points) How many **distinct** irreducible permutation representations of $\mathfrak{S}_3$ are there? Take into account that the subgroups of $\mathfrak{S}_3$ are $H_1 = \{e\}$, $H_2 = \langle(12)\rangle$, $H_3 = \langle(13)\rangle$, $H_4 = \langle(23)\rangle$, $H_5 = \langle(123)\rangle$, and $H_6 = \mathfrak{S}_3$.

Solution: By the correspondence between subgroups and transitive permutation representations, the irreducible permutation representations of $\mathfrak{S}_3$ correspond to the conjugacy classes of subgroups. The distinct irreducible permutation representations are obtained from the subgroups:

- $H_1 = \{e\}$ gives the regular representation of cardinality $6 = |\mathfrak{S}_3|$.
- $H_5 = \langle(123)\rangle$ of order 3 gives a transitive representation of cardinality $[S_3 : H_5] = 2$.
- $H_2 = \langle(12)\rangle$ of order 2 gives a transitive representation of cardinality $[S_3 : H_2] = 3$.
- $H_6 = \mathfrak{S}_3$ gives the trivial representation of cardinality 1.

The conjugacy classes of subgroups of order 2 are all equivalent (since all involutions are conjugate in $\mathfrak{S}_3$). Therefore, there are exactly 4 distinct irreducible permutation representations of $\mathfrak{S}_3$.

2. (5 points) Suppose that $\mathcal{C}$ is a family of subsets of a group G which forms a partition of G , and suppose further that $gC \in \mathcal{C}$ for any $g \in G$ and $C \in \mathcal{C}$. Show that there is an $H \in \mathcal{C}$ which is a subgroup of G . (Note that this is only part of the proof to show that $\mathcal{C}$ is a set of left cosets.)

Solution: Let H denote the unique element of $\mathcal{C}$ that contains the identity e (such an element exists because $\mathcal{C}$ partitions G). We will show that H is a subgroup of G and that $\mathcal{C} = \{gH : g \in G\}$.

Closure: Let $h_1, h_2 \in H$. Consider the set $h_1H \in \mathcal{C}$. Since $h_1 = h_1 \cdot e \in h_1H$ and $h_1 \in H$, the two members h_1H and H of the partition $\mathcal{C}$ share the element h_1 . Because $\mathcal{C}$ is a partition (its members are pairwise disjoint), we must have $h_1H = H$. In particular, $h_1h_2 \in h_1H = H$.

Identity: $e \in H$ by definition.

Inverses: Let $h \in H$. From the closure argument we know $hH = H$. Since $e \in H$, there exists $h' \in H$ such that $hh' = e$, i.e., $h^{-1} = h' \in H$.

Therefore $H \leq G$.

3. (5 points) Assuming that we already know that $\text{Inn}(G) \leq \text{Aut}(G)$ for any group G , show that $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

Solution: Let $\varphi_g : G \rightarrow G$ be the inner automorphism defined by $\varphi_g(x) = gxg^{-1}$ for any $g \in G$. Then $\text{Inn}(G) = \{\varphi_g : g \in G\}$.

To show that $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$, we must show that for any $\varphi_g \in \text{Inn}(G)$ and any $\sigma \in \text{Aut}(G)$, the conjugate $\sigma \circ \varphi_g \circ \sigma^{-1}$ is also in $\text{Inn}(G)$.

Let $\sigma \in \text{Aut}(G)$ and $g \in G$. Then for every $x \in G$,

$$(\sigma \circ \varphi_g \circ \sigma^{-1})(x) = \sigma(\varphi_g(\sigma^{-1}(x))) = \sigma(g\sigma^{-1}(x)g^{-1}) = \sigma(g) \cdot x \cdot (\sigma(g))^{-1} = \varphi_{\sigma(g)}(x).$$

So $\sigma \circ \varphi_g \circ \sigma^{-1} = \varphi_{\sigma(g)}$, which is an inner automorphism since $\sigma(g) \in G$. Therefore, $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

4. (5 points) Let G be a finite group and let H be a subgroup of G with $[G : H] = p$, where p is the smallest prime dividing $|G|$. Prove that H is normal in G .

Solution: Let G act on the set G/H of left cosets by left multiplication, i.e., $g \cdot (aH) = (ga)H$. This gives a homomorphism $\phi : G \rightarrow \mathfrak{S}_{G/H} \cong \mathfrak{S}_p$, where $\mathfrak{S}_p$ is the symmetric group on p elements.

Let $K = \ker(\phi)$. Since K is the kernel of a homomorphism, $K \trianglelefteq G$. We also know that $K \subseteq H$: indeed, if $g \in K$ then g fixes every coset, and in particular $g \cdot (eH) = gH = H$, which means $g \in H$. So $K \leq H \leq G$.

By the First Isomorphism Theorem, $G/K \cong \text{im}(\phi) \leq \mathfrak{S}_p$, so $|G/K|$ divides $|\mathfrak{S}_p| = p!$. That is, $[G : K]$ divides $p!$.

On the other hand, by the tower law for indices,

$$[G : K] = [G : H] \cdot [H : K] = p \cdot [H : K].$$

Now $[G : K]$ divides $|G|$ (since $K \leq G$) and $[G : K]$ divides $p!$. Since p is the *smallest* prime dividing $|G|$, this can only happen if $[G : K]$ divides p , and so $[G : K]$ is either 1 or p . If $[G : K] = 1$, then $K = G$, which contradicts the fact that $K \subseteq H$ and $[G : H] = p > 1$. Therefore, we must have $[G : K] = p$ and $[H : K] = 1$, i.e., $K = H$.

Since $K = \ker(\phi) \trianglelefteq G$, we conclude $H \trianglelefteq G$.

5. (5 points) **(Cayley's Theorem)** Prove that every group G is isomorphic to a subgroup of the symmetric group $\mathfrak{S}_G$ (the group of all bijections $G \rightarrow G$). In particular, if G is finite with $|G| = n$, then G is isomorphic to a subgroup of $\mathfrak{S}_n$.

Solution: Consider the left regular action of G on itself where G acts by left multiplication. This defines a group homomorphism $\phi : G \rightarrow \mathfrak{S}_G$ where for every $g \in G$ and every $x \in G$, $\phi(g)(x) = gx$. The kernel of ϕ is given by

$$\begin{aligned}\ker(\phi) &= \{g \in G : \phi(g)(x) = x \text{ for all } x \in G\} \\ &= \{g \in G : gx = x \text{ for all } x \in G\} \\ &= \{g \in G : g = e\} \\ &= \{e\}.\end{aligned}$$

We conclude that ϕ is injective, and by the First Isomorphism Theorem,

$$G \cong G/\{e\} = G/\ker(\phi) \cong \text{im}(\phi) \leq \mathfrak{S}_G.$$

If $|G| = n$, we can identify $\mathfrak{S}_G$ with $\mathfrak{S}_n$, and thus G is isomorphic to a subgroup of $\mathfrak{S}_n$.

6. (5 points) Let H and K be subgroups of a group G . Prove that if the product set HK is a subgroup of G then $HK = KH$. (Note that this is only one direction)

Solution: Recall $HK = \{hk : h \in H, k \in K\}$ and $KH = \{kh : k \in K, h \in H\}$.

Assume HK is a subgroup of G . We show $HK = KH$.

($\subseteq$) Let $hk \in HK$. Since $HK \leq G$, we have $(hk)^{-1} = k^{-1}h^{-1} \in HK$, so $k^{-1}h^{-1} = h'k'$ for some $h' \in H, k' \in K$. Taking inverses of both sides: $hk = (k')^{-1}(h')^{-1} \in KH$. Hence $HK \subseteq KH$.

($\supseteq$) Let $kh \in KH$. Then $(kh)^{-1} = h^{-1}k^{-1} \in HK$. Since HK is a subgroup, $((kh)^{-1})^{-1} = kh \in HK$. Hence $KH \subseteq HK$.

Therefore $HK = KH$.